

Higher-Genus Zamolodchikov Recursion for $\mathcal{N} = 1$ Superconformal Blocks

Representation-theory kernel and plumbing-graph assembly

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Abstract

We isolate the representation-theory kernel required for a numerical higher-genus $\mathcal{N} = 1$ superconformal-block recursion. As in the Virasoro construction of Cho–Collier–Yin, every internal edge contributes three local factors: a pole-conversion Jacobian J_{rs} , an inverse null-norm residue A_{rs} , and one fusion polynomial (or Ramond fusion matrix) at each end of the edge. We derive these factors for NS modules and for generic long Ramond modules, including the G_0 ground-state doublet and its shortening limit. The central-charge recursion holds NS weights but Ramond momenta β , not Ramond absolute weights, fixed; this keeps $G_0^2 = -\beta^2$ constant and removes the spurious level-zero central-charge pole. The sphere four-point and torus one-point recursions are used as local calibrations, not as topology-specific starting points. Once these local residues are fixed, the genus- g polar recursion is obtained by inserting the same kernel on every decorated edge of a trivalent plumbing graph. The remaining topology-sensitive input is the large-central-charge seed and the supergeometric sewing ledger (spin lift, odd plumbing data, and defect-line routing).

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1 Purpose, scope, and recommended route

The goal is a numerical chiral-block engine for a punctured genus- g super-Riemann surface in a chosen degeneration channel. The derivation is a supersymmetric adaptation of Ref. [1]. Its local input is the NS and Ramond representation theory worked out in Refs. [2, 3, 4, 5, 6]. The block is first defined algebraically by ordinary state sewing. The super-Riemann-surface description then fixes the allowed NS/R edge labels, the lift of every plumbing parameter, the odd moduli, and the spin-structure signs. This separation is useful numerically: representation-theory kernels remain reusable while the geometric ledger is kept explicit.

The work is naturally divided as follows:

derive J_{rs} , A_{rs} , and P_{rs} from NS/R modules, calibrate this local kernel against sphere and torus recursion, assemble it on an arbitrary decorated plumbing graph, treat the Ramond shortening point as a separate boundary case.	(1)
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The first three steps concern generic Verma modules and are topology independent. The higher-genus problem begins only after them: ordinary CFT sewing computes a block after the supergeometric choices have been supplied. The generic long-R block is matrix-valued in its ground-state

indices. A short Ramond ground state is obtained by projection to the irreducible quotient, not by naively substituting $h = c/24$ in a normalized long basis.

The present note derives the holomorphic block. A type-0 string integrand requires, in addition, the antiholomorphic block, the GSO sum, superghosts, the supermoduli measure, picture-changing or an equivalent supergeometric prescription, and the Liouville structure constants. None of these is silently included in $\mathcal{F}$ below.

2 Central-charge and weight conventions

We use the ordinary central charge throughout:

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}, \quad (2)$$

$$\begin{aligned} [L_m, G_r] &= \left(\frac{m}{2} - r\right) G_{m+r}, \\ \{G_r, G_s\} &= 2L_{r+s} + \frac{c}{3}\left(r^2 - \frac{1}{4}\right)\delta_{r+s,0}, \quad r, s \in \mathbb{Z} + \frac{1}{2}. \end{aligned} \quad (3)$$

For super-Liouville theory,

$$Q = b + b^{-1}, \quad c = \frac{3}{2}(1 + 2Q^2), \quad h(\lambda) = \frac{Q^2 - \lambda^2}{8}. \quad (4)$$

Thus the type-0B point $b = 1$ has $c = 27/2$.

2.1 BRY–HJS convention lock

The sphere recursion with Ramond insertions in Ref. [3] uses S_n for the supercurrent modes and Δ for the conformal weight. In the notation used by Xi Yin and BRY the exact dictionary is

$$S_n^{\text{HJS}} = G_n^{\text{BRY}}, \quad \Delta^{\text{HJS}} = h^{\text{BRY}}, \quad c^{\text{HJS}} = c^{\text{BRY}}. \quad (5)$$

In particular, the HJS paper uses the ordinary central charge in

$$[L_m, S_n] = \left(\frac{m}{2} - n\right) S_{m+n}, \quad \{S_m, S_n\} = 2L_{m+n} + \frac{c}{3}\left(m^2 - \frac{1}{4}\right)\delta_{m+n,0}, \quad (6)$$

so no $c \leftrightarrow \hat{c}$ rescaling is made when its four-point formulas are imported. This is distinct from the Belavin–Geiko convention described below.

For the symmetric Ramond basis of HJS,

$$S(z)R_\beta^\pm(u) \sim i\beta e^{\mp i\pi/4}(z - u)^{-3/2}R_\beta^\mp(u), \quad h_R(\beta) = \frac{c}{24} - \beta^2. \quad (7)$$

The BRY super-Liouville momentum is therefore

$$\beta = \frac{iP}{\sqrt{2}}, \quad h_R(P) = \frac{c}{24} + \frac{P^2}{2}, \quad (8)$$

with no extra factor of $\sqrt{2}$ in P . The signs attached to β in an HJS chiral three-point form label chiral Ward-identity solutions. They must not be identified with the BRY nonchiral families $V^{\text{R},+}$ and $V^{\text{R},-}$. The latter distinction is made only after holomorphic and antiholomorphic blocks are combined with the physical three-point constants and defect-line phases.

Belavin–Geiko and several superconformal-recursion references instead call

$$\widehat{c} \equiv \frac{2c}{3} = 1 + 2Q^2 \quad (9)$$

the central charge. Their algebra has central terms $\widehat{c}/8$ and $\widehat{c}/2$ in Eqs. (2) and (3). Hence $\widehat{c} = 9$, not $27/2$, at $b = 1$. All pole denominators and Jacobians must use the same convention. If $J_{rs} = -\partial c_{rs}/\partial h$, then

$$J_{rs} = \frac{3}{2}\widehat{J}_{rs}, \quad c - c_{rs} = \frac{3}{2}(\widehat{c} - \widehat{c}_{rs}), \quad (10)$$

so the ratio $J_{rs}/(c - c_{rs})$ is convention-independent.

An NS Verma module $\mathcal{H}_h$ is spanned by

$$\mathcal{L}_I \nu_h = G_{-r_1} \cdots G_{-r_p} L_{-n_1} \cdots L_{-n_k} \nu_h, \quad |I| = \sum_a r_a + \sum_b n_b \in \frac{1}{2}\mathbb{Z}_{\geq 0}, \quad (11)$$

with strictly ordered fermionic modes. We normalize $\langle \nu_h | \nu_h \rangle = 1$ and denote the level- N Gram matrix by

$$B_{h,c}^{(N)}[I, J] = \langle \nu_h | \mathcal{L}_I^\dagger \mathcal{L}_J | \nu_h \rangle. \quad (12)$$

There are two independent normalized NS three-point forms,

$$\rho^0(\nu_1, \nu_2, \nu_3) = 1, \quad \rho^1(\nu_1, \nu_2, \nu_3) \equiv \rho(\nu_1, G_{-1/2}\nu_2, \nu_3) = 1, \quad (13)$$

where the second equality in each definition fixes the kinematic normalization; physical even and odd structure constants multiply these forms later. A component label $\alpha_v \in \{0, 1\}$ must therefore be stored at every trivalent NS vertex.

For a Ramond module the fermionic modes are integral and

$$G_0^2 = L_0 - \frac{c}{24}. \quad (14)$$

It is useful to parameterize a Ramond weight by

$$h_R(\beta) = \frac{c}{24} - \beta^2, \quad \beta = \frac{iP}{\sqrt{2}} \quad \text{for a Liouville momentum } P, \quad (15)$$

and to choose parity eigenstates $w_\beta^\pm$ obeying

$$(-1)^F w_\beta^\pm = \pm w_\beta^\pm, \quad G_0 w_\beta^\pm = i e^{\mp i\pi/4} \beta w_\beta^\mp. \quad (16)$$

For $\beta \neq 0$ this is a generic long ground-state doublet. An especially safe symbolic basis is

$$(w^+, G_0 w^+), \quad B_R^{(0)} = \text{diag} \left(1, h - \frac{c}{24} \right), \quad (17)$$

up to the fixed BPZ phase convention. It keeps every matrix entry rational and displays the shortening zero. A unit-normalized long basis is obtained only for $\kappa^2 = h - c/24 \neq 0$ by multiplying the second state by κ^{-1} . At $\kappa = 0$, $G_0 w^+$ is null and the irreducible short module retains one ground state. Thus

$$\det B_R^{(0)} = h - \frac{c}{24} \quad (18)$$

is a level-zero Kac zero in addition to the positive-level zeros discussed below. This is the only genuinely new representation-theory subtlety relative to an NS edge.

3 The block in a plumbing frame

Let Γ be a trivalent pants graph for a genus- g surface with N punctures. It has $3g - 3 + N$ internal edges and $2g - 2 + N$ vertices. Each internal edge carries a sector $s_e \in \{\text{NS}, \text{R}\}$, a weight h_e , a plumbing parameter q_e , and two descendant labels I_e, J_e of equal level n_e . With the primary powers $\prod_e q_e^{h_e}$ stripped off, define

$$\mathcal{F}_\Gamma^\alpha(c, h; q) = \sum_{n \in (\frac{1}{2}\mathbb{Z}_{\geq 0})^{E(\Gamma)}} \prod_e q_e^{n_e} \sum_{\substack{|I_e|=|J_e|=n_e \\ e \in E(\Gamma)}} \times \left[\prod_e (B_{s_e, h_e, c}^{(n_e)})^{-1}[I_e, J_e] \right] \left[\prod_v \rho_v^{\alpha_v}(\{\mathcal{L}_{K(v, e)} \nu_{h_e}\}) \right]. \quad (19)$$

Here $K(v, e)$ is I_e or J_e according to the chosen orientation of e ; external legs are replaced by their fixed states. All three-point forms are plane forms at $0, 1, \infty$, and all boundaries are sewn by global superconformal maps. This choice is the super-analogue of the CCY plumbing frame. A different local-coordinate frame generally multiplies the block by a conformal-anomaly factor and must not be mixed with Eq. (19).

For an NS edge, $n_e \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ and the ground fiber is one-dimensional. For an R edge, $n_e \in \mathbb{Z}_{\geq 0}$ and the descendant label includes a ground index $a_e = \pm$. Before sewing the two ends, the block is therefore a tensor in the open Ramond ground fibers

$$\mathcal{G}_{h_e} = \text{span}\{w_{h_e}^+, w_{h_e}^-\}. \quad (20)$$

The BPZ/Siegel pairing contracts these indices only after the local three-point tensors have been multiplied. In particular, the matrix size does not grow under recursion: every open Ramond line always carries a two-dimensional fiber. On an allowed pants graph, R edges form paths or cycles through NS–R–R vertices, so their ground indices are multiplied along the corresponding fermion-parity defect lines.

For example, the all-NS genus-two theta graph has three internal edges:

$$\mathcal{F}_\Theta^{\alpha_L, \alpha_R} = \sum_{n_1, n_2, n_3} \prod_{i=1}^3 q_i^{n_i} \sum_{\substack{|I_i|=|J_i|=n_i \\ i=1,2,3}} \prod_{i=1}^3 (B_{h_i, c}^{(n_i)})^{-1}[I_i, J_i] \times \rho^{\alpha_L}(\mathcal{L}_{I_1} \nu_{h_1}, \mathcal{L}_{I_2} \nu_{h_2}, \mathcal{L}_{I_3} \nu_{h_3}) \rho^{\alpha_R}(\mathcal{L}_{J_1} \nu_{h_1}, \mathcal{L}_{J_2} \nu_{h_2}, \mathcal{L}_{J_3} \nu_{h_3}). \quad (21)$$

Equation (21), not a conjectured closed product, is the reference definition for low-level numerical tests.

4 Universal representation-theory residue data

4.1 Where the factors $J_{rs} A_{rs} P_{rs} P_{rs}$ come from

The recursion coefficient is not an ansatz. It is the residue of one inverse Gram matrix. Let a singular vector at level $N_{rs} = rs/2$ be normalized by

$$\chi_{rs}(c, h) = D_{rs}(c, h) \nu_h, \quad D_{rs} = L_{-1}^{N_{rs}} + \cdots, \quad (22)$$

with the appropriate fermionic leading monomial when necessary. Near a Kac zero $h = h_{rs}(c)$ its norm is

$$\langle \chi_{rs} | \chi_{rs} \rangle = \frac{h - h_{rs}(c)}{A_{rs}^{(h)}(c)} + O((h - h_{rs})^2). \quad (23)$$

Thus $A_{rs}^{(h)}$ is the inverse slope of the null norm. If the recursion variable is c at fixed h , then

$$h - h_{rs}(c) = - \left. \frac{\partial h_{rs}}{\partial c} \right|_{c_{rs}} (c - c_{rs}) + \cdots, \quad J_{rs}(h) \equiv - \frac{\partial c_{rs}(h)}{\partial h}, \quad (24)$$

and the inverse-Gram residue contains $J_{rs}A_{rs}^{(h)}$. This is the factor $J_{rs}A_{rs}$.

The remaining factors come from decoupling the null submodule at the two vertices adjacent to the singular edge. Super-Virasoro Ward identities give

$$\rho_v(\dots, D_{rs}\nu_h, \dots) = P_{rs}^{(v,e)} \rho_v(\dots, \nu_{h+N_{rs}}, \dots). \quad (25)$$

The coefficient $P_{rs}^{(v,e)}$ is a scalar for a purely NS vertex and a linear map on Ramond ground fibers for an NS-R-R vertex. Applying Eq. (25) at the two ends of an edge produces the two factors conventionally denoted $P_{rs}P_{rs}$. Consequently,

$$\boxed{\text{edge residue} = (\text{pole Jacobian}) (\text{inverse null-norm slope}) (\text{left fusion map}) (\text{right fusion map}).} \quad (26)$$

This representation-theory statement is independent of genus and is the main reusable result of the sphere recursion literature.

4.2 NS Kac zeros and pole locations

The NS module is degenerate when

$$\lambda_{r,s} = rb + sb^{-1}, \quad r, s \in \mathbb{Z}_{>0}, \quad r + s \in 2\mathbb{Z}, \quad (27)$$

and then contains a null vector $\chi_{r,s}$ at level $rs/2$. At fixed internal weight h , the pole $c = c_{rs}(h)$ is found from

$$h = \frac{(b + b^{-1})^2 - (rb + sb^{-1})^2}{8}. \quad (28)$$

Writing $x = b^2$, this is the quadratic equation

$$(r^2 - 1)x^2 + [8h + 2(rs - 1)]x + (s^2 - 1) = 0. \quad (29)$$

The branch used by the present sphere code is

$$x_{rs}(h) = - \frac{4h + rs - 1 + \sqrt{16h^2 + 8(rs - 1)h + (r - s)^2}}{r^2 - 1}, \quad (30)$$

$$c_{rs}(h) = \frac{3}{2} \left[1 + 2 \left(b_{rs} + b_{rs}^{-1} \right)^2 \right] = \frac{15}{2} + 3x_{rs} + \frac{3}{x_{rs}}, \quad b_{rs} = \sqrt{x_{rs}}. \quad (31)$$

Both square roots are principal initially and then continued continuously in h . A numerical implementation must cache the chosen branch rather than reselect it independently at nearby momenta. Define

$$J_{rs}(h) = - \frac{\partial c_{rs}(h)}{\partial h}. \quad (32)$$

4.3 Null norm and fusion polynomials

With the singular vector normalized as in Ref. [6], the inverse-norm residue is

$$A_{rs}(h) = \frac{1}{2} \prod_{\substack{p=1-r, \dots, r \\ q=1-s, \dots, s \\ p+q \in 2\mathbb{Z} \\ (p,q) \neq (0,0), (r,s)}} \left[\frac{pb_{rs} + qb_{rs}^{-1}}{\sqrt{2}} \right]^{-1}. \quad (33)$$

For any weight d_i , define λ_i at the pole by

$$d_i = \frac{(b_{rs} + b_{rs}^{-1})^2 - \lambda_i^2}{8}, \quad (34)$$

with a continuous square-root choice. The two fusion polynomials are

$$P_{rs}^\alpha(d_i, d_j; h) = \prod_{p=1-r, \text{ step } 2}^{r-1} \prod_{q=1-s, \text{ step } 2}^{s-1} \left(\frac{\lambda_i - \lambda_j + pb_{rs} + qb_{rs}^{-1}}{2\sqrt{2}} \right) \left(\frac{\lambda_i + \lambda_j + pb_{rs} + qb_{rs}^{-1}}{2\sqrt{2}} \right), \quad (35)$$

where the product retains

$$p + q - (r + s) \in \begin{cases} 4\mathbb{Z} + 2, & \alpha = 0, \\ 4\mathbb{Z}, & \alpha = 1. \end{cases} \quad (36)$$

The null vector changes parity when rs is odd, so the factor entering a three-point form is

$$\sigma_{rs}^\alpha = \begin{cases} P_{rs}^\alpha, & rs \in 2\mathbb{Z}, \\ P_{rs}^{1-\alpha}, & rs \in 2\mathbb{Z} + 1. \end{cases} \quad (37)$$

Reversing the null leg introduces

$$\rho^\alpha(\chi_{rs}, \nu_i, \nu_j) = \sigma_{rs}^\alpha(d_i, d_j; h), \quad (38)$$

$$\rho^\alpha(\nu_i, \nu_j, \chi_{rs}) = S_{rs}^\alpha \sigma_{rs}^\alpha(d_j, d_i; h), \quad S_{rs}^0 = 1, \quad S_{rs}^1 = (-1)^{rs}. \quad (39)$$

Equations (38)–(39) are the orientation ledger. A code should first rotate each trivalent tensor to one of these canonical orderings; inserting signs ad hoc after multiplication is unsafe.

4.4 Ramond Kac zeros and the shifted long module

A positive-level Ramond singular vector occurs for

$$r, s \in \mathbb{Z}_{>0}, \quad r + s \in 2\mathbb{Z} + 1, \quad \beta_{rs} = \frac{rb + sb^{-1}}{2\sqrt{2}}, \quad (40)$$

at the integer level $N_{rs} = rs/2$. Its degenerate weight is

$$h_{rs}^R(c) = \frac{c}{24} - \beta_{rs}^2. \quad (41)$$

Choose the positive-parity singular vector

$$\chi_{rs}^+ = D_{rs} w_{\beta_{rs}}^+, \quad D_{rs} = L_{-1}^{rs/2} + \dots. \quad (42)$$

Acting with G_0 supplies its odd partner. Both generate the same shifted long Ramond module, whose momentum parameter is

$$\beta'_{rs} = \frac{(-1)^s}{2\sqrt{2}} \left(rb - \frac{s}{b} \right), \quad h_R(\beta'_{rs}) = h_{rs}^R + \frac{rs}{2}. \quad (43)$$

This identity is the Ramond version of the familiar weight shift under a null submodule. It also shows why the residue acts on a two-component block: the null module is again a generic long R module.

Correction: the Ramond c -recursion is taken at fixed β , not fixed h_R . The earlier fixed-weight slice makes $G_0^2 = h_R - c/24$ and the Ramond ground representation vary with c . It also introduces an avoidable level-zero pole at $c = 24h_R$. Instead write

$$h_R(c, \beta) = \frac{c}{24} - \beta^2 \quad (44)$$

and hold β fixed. Then both $G_0^2 = -\beta^2$ and the primary propagator $q^{h_R - c/24} = q^{-\beta^2}$ are independent of c .

The pole $c_{rs}^R(\beta; \epsilon)$ is defined by $\beta = \epsilon\beta_{rs}(b_*)$, $\epsilon = \pm 1$. Equivalently,

$$rb_*^2 - \epsilon 2\sqrt{2} \beta b_* + s = 0, \quad (45)$$

and

$$c_{rs}^R(\beta; \epsilon) = \frac{3}{2} + 3 \left(b_* + b_*^{-1} \right)^2. \quad (46)$$

The quadratic branches, together with $(r, s, b) \leftrightarrow (s, r, b^{-1})$, must be organized before equal c -poles are merged.

Along this fixed- β slice,

$$h_R(c, \beta) - h_{rs}^R(c) = \beta_{rs}(c)^2 - \beta^2. \quad (47)$$

Thus the weight-residue Jacobian is

$$\begin{aligned} J_{rs}^R(\beta; \epsilon) &= \left[\partial_c \beta_{rs}(c)^2 \right]_{c=c_{rs}^R(\beta; \epsilon)}^{-1} \\ &= \frac{24(b + b^{-1})(1 - b^{-2})}{(rb + sb^{-1})(r - sb^{-2})} \Big|_{b=b_*}. \end{aligned} \quad (48)$$

Equivalently, importing the published β -residue normalization $A_{rs}^{R,(\beta)} = -A_{rs}^{R,(h)}/(2\beta_{rs})$, the complete pullback factor is

$$J_{rs}^R A_{rs}^{R,(h)} = - \frac{A_{rs}^{R,(\beta)}}{\partial_c \beta_{rs}} \Big|_{c=c_{rs}^R(\beta; \epsilon)}. \quad (49)$$

No new Ramond singular-vector calculation is required.

4.5 The first Ramond Kac zero as a unit test

The lowest positive-level R degeneracies are $(r, s) = (2, 1)$ and $(1, 2)$ at level one. They are exchanged by $b \leftrightarrow b^{-1}$. In the unnormalized positive-parity basis

$$u_1 = L_{-1}w^+, \quad u_2 = G_{-1}G_0w^+, \quad \kappa^2 = h - \frac{c}{24}, \quad (50)$$

the algebra gives, up to the already fixed BPZ phase,

$$B_{\text{R}}^{(1,+)} = \begin{pmatrix} 2h & \frac{3}{2}\kappa^2 \\ \frac{3}{2}\kappa^2 & \kappa^2(2h + \frac{c}{4}) \end{pmatrix}. \quad (51)$$

Its determinant separates the two different mechanisms:

$$\det B_{\text{R}}^{(1,+)} = \kappa^2 \left[2h \left(2h + \frac{c}{4} \right) - \frac{9}{4}\kappa^2 \right]. \quad (52)$$

The first factor is inherited from ground-state shortening. The two roots of the bracket are precisely the $(2, 1)$ and $(1, 2)$ Kac weights,

$$32h^2 + (4c - 18)h + \frac{3c}{4} = 0. \quad (53)$$

On either branch the normalized singular operator can be written

$$D_{(1)} = L_{-1} - \frac{6}{8h + c} G_{-1}G_0 = L_{-1} - \frac{4h}{3\kappa^2} G_{-1}G_0. \quad (54)$$

For the fixed- β c -recursion one branch is selected by the chosen solution of Eq. (45). Equations (51)–(54) are the cheapest nontrivial test of the R algebra layer: the code must distinguish the shortening factor from the positive-level residue before evaluating any fusion polynomial.

For the mixed ordering $\langle \phi_4 R_3 R_2 \phi_1 \rangle$ let $\eta_3, \eta_2 = \pm 1$ label the two HJS chiral forms. The level-one three-form vectors in the basis (50) are

$$\begin{aligned} \ell^{(\eta_3)} &= \left(h + h_3 - h_4, \frac{\kappa^2}{2} - \eta_3 \beta \beta_3 \right), \\ r^{(\eta_2)} &= \left(h + h_2 - h_1, \frac{\kappa^2}{2} - \eta_2 \beta \beta_2 \right). \end{aligned} \quad (55)$$

The G_0 phases in the two slots appear different before BPZ conjugation; the anti-linearity of the bra three-form makes the two displayed components identical in form. Thus the direct, nonrecursive local coefficient is

$$F_{\eta_3, \eta_2}^{(1)} = \ell^{(\eta_3)} \left(B_{\text{R}}^{(1,+)} \right)^{-1} r^{(\eta_2)}. \quad (56)$$

This formula is the brute-force unit test used below; it contains neither a Kac product nor an elliptic recursion.

4.6 What the present G_0 test fixes

There are two logically different Ramond sphere calculations in the code. The four-R block propagates an NS module, so its internal inverse Gram matrix contains no G_0 . Its Ramond data live only in the two external N–R–R three-point forms. By contrast, the crossed RRNSNS block propagates a generic long-R module and therefore genuinely tests the ground doublet. In the ordered basis (w^+, w^-) the implementation keeps

$$G_0 = \begin{pmatrix} 0 & ie^{i\pi/4}\beta \\ ie^{-i\pi/4}\beta & 0 \end{pmatrix}, \quad G_0^2 = -\beta^2 \mathbf{1} = \left(h - \frac{c}{24} \right) \mathbf{1}. \quad (57)$$

For all four pairs (η_3, η_2) , the recursive level-one coefficient agrees with the independent contraction (56). The same agreement now holds after the exact self-dual finite-part projection at $c = 27/2$; for the representative choice

$$(P_1^{\text{NS}}, P_2^{\text{R}}, P_3^{\text{R}}, P_4^{\text{NS}}; P) = (0.31, 0.41, 0.23, 0.37; 0.70), \quad (\eta_3, \eta_2) = (+, -), \quad (58)$$

both methods give

$$F_{+,-}^{(1)} = 0.461591230220044. \quad (59)$$

Thus there is no outstanding G_0 sign or phase ambiguity for this scalar sphere component.

Two qualifications remain. First, $\beta = 0$ is not a generic long representation: $G_0 w^+$ becomes null and the ground Gram matrix shortens. The long-R code now rejects $P = 0$ explicitly; the irreducible short quotient must be constructed separately. This causes no endpoint problem in the present continuum quadrature, whose Gauss nodes lie at $P > 0$. Second, the sphere class contracts the ground indices into one HJS sign component. A higher-genus R edge must instead keep the full 2×2 fiber open, insert the BPZ/odd-plumbing map in a fixed order, and only then apply the GSO contraction. The remaining risk is therefore matrix routing around an R cycle, not the local algebraic action of G_0 .

4.7 Brute-force Ramond descendants through q^3

The level-one comparison can be extended without using any Kac residue. In the unnormalized ground basis, the positive-parity PBW basis at integer level N is

$$\mathcal{B}_N^+ = \left\{ L_{-\lambda} G_{-\mu} G_0^\epsilon w^+ : |\lambda| + |\mu| = N, \quad \mu \text{ strict}, \quad \ell(\mu) + \epsilon \in 2\mathbb{Z} \right\}. \quad (60)$$

Its dimensions through level three are

$$\dim \mathcal{B}_{0,1,2,3}^+ = 1, 2, 4, 8. \quad (61)$$

The code normal-orders $\text{BPZ}(u_i)u_j$ using the ordinary- c algebra and therefore constructs every Gram entry independently of the determinant formula.

The N–R–R Ward identities also give a triangular direct evaluator. If $\mathcal{V}_a(u)$ denotes a normalized three-form with external Ramond ground label $a = \pm$, then

$$\begin{aligned} \mathcal{V}_a(L_{-n}u) &= (h + N(u) + nh_{\text{R}} - h_{\text{NS}}) \mathcal{V}_a(u), \\ \mathcal{V}_a(G_{-m}u) &= \frac{g_a \mathcal{V}_a(u)}{i(-1)^{p(u)}} - \sum_{p \geq 1} \binom{1/2}{p} (-1)^p \mathcal{V}_a(G_{-m+p}u), \end{aligned} \quad (62)$$

where

$$N(u) = \text{level of } u, \quad p(u) = \text{fermion parity of } u, \quad g_+ = ie^{-i\pi/4} \beta_{\text{ext}}, \quad g_- = ie^{i\pi/4} \beta_{\text{ext}}. \quad (63)$$

The sum truncates because every $p > 0$ lowers the total descendant level. The HJS sign convention used by the recursion is fixed by

$$\mathcal{V}_+(w^+) = 1, \quad \mathcal{V}_-(G_0 w^+) = -\eta ie^{-i\pi/4} \beta. \quad (64)$$

Equation (62) is Eq. (26) of Ref. [4] specialized to primary external states at $z = 1$. The RN form on the other pair of pants has the identical algebraic form after BPZ conjugation.

At each level the local coefficient is then computed only from

$$F_N^{\text{direct}} = \ell_N^T \left(B_R^{(N,+)} \right)^{-1} \mathbf{r}_N. \quad (65)$$

Expanding the known elliptic prefactor converts $1 + \sum_{N=1}^3 F_N^{\text{direct}} z^N$ to $H(q)$ through q^3 . For

$$(P_1^{\text{NS}}, P_2^{\text{R}}, P_3^{\text{R}}, P_4^{\text{NS}}; P) = (0.31, 0.41, 0.23, 0.37; 0.70) \quad (66)$$

at the exact self-dual point, the result is

(η_3, η_2)	$[q]H$	$[q^2]H$	$[q^3]H$
$(+, +)$	-0.467734839362781	0.376214080317261	-0.532846461739354
$(+, -)$	-0.526540316479303	0.330524749436549	-0.605488031080000
$(-, +)$	-0.603070765624170	0.346914586975835	-0.576402689452465
$(-, -)$	-0.012744163247911	0.189125920915153	-0.570139990696305

Every entry agrees with the collision-aware recursion; the largest absolute difference is 8.77×10^{-13} , attained in the $(+, -)$ q^3 coefficient. This comparison tests the G_0 action, all Gram matrices, both Ward vectors, BPZ ordering, and the local- z to elliptic- q conversion simultaneously. It is substantially stronger than the original level-one check.

4.8 Ramond inverse norm and fusion polynomials

In the normalization (42), the inverse slope with respect to the conformal weight is

$$A_{rs}^{\text{R},(h)}(c) = 2^{rs-2} \left(rb + \frac{s}{b} \right) \prod_{\substack{m=1-r, \dots, r \\ n=1-s, \dots, s \\ m+n \in 2\mathbb{Z} \\ (m,n) \neq (0,0)}} \left(mb + \frac{n}{b} \right)^{-1}. \quad (67)$$

This is the factor appropriate to a c -recursion through $J_{rs}^{\text{R}} A_{rs}^{\text{R},(h)}$. Sphere formulas are often written instead as a recursion in β . Since

$$h_{\text{R}}(\beta) - h_{rs}^{\text{R}} = -(\beta - \beta_{rs})(\beta + \beta_{rs}), \quad (68)$$

the residue at the positive β pole contains

$$A_{rs}^{\text{R},(\beta)} = -\frac{A_{rs}^{\text{R},(h)}}{2\beta_{rs}} = -2^{rs-3/2} \prod_{\substack{m=1-r, \dots, r \\ n=1-s, \dots, s \\ m+n \in 2\mathbb{Z} \\ (m,n) \neq (0,0)}} \left(mb + \frac{n}{b} \right)^{-1}, \quad (69)$$

where (r, s) is absent automatically because $r + s$ is odd. The pole at $-\beta_{rs}$ has the opposite Jacobian sign. The odd-sublattice condition printed beneath the corresponding product in Ref. [4] is inconsistent with the direct level-one Gram residue. The even sublattice above reproduces both β_{21} and β_{12} residues and is independently selected by the mixed-channel crossing test below. Keeping $A^{\text{R},(h)}$ and $A^{\text{R},(\beta)}$ as distinct code objects prevents a common factor-of-2 β_{rs} error.

The N–R–R null-vector matrix elements are encoded by two Ramond fusion polynomials. Write the NS weight as $d = h(\lambda)$ and first define

$$\begin{aligned} X_e^{rs}(u) &= \prod_{\substack{0 \leq k \leq r-1, \ 0 \leq l \leq s-1 \\ k+l \in 2\mathbb{Z}}} \frac{u - (1-r+2k)b - (1-s+2l)b^{-1}}{2\sqrt{2}}, \\ X_o^{rs}(u) &= \prod_{\substack{0 \leq k \leq r-1, \ 0 \leq l \leq s-1 \\ k+l \in 2\mathbb{Z}+1}} \frac{u - (1-r+2k)b - (1-s+2l)b^{-1}}{2\sqrt{2}}. \end{aligned} \quad (70)$$

For a Ramond leg of parameter β , the convention used in Refs. [4, 5] is

$$P_{rs}^{\text{R},\pm}(d, \beta; c) = X_e^{rs}(2\sqrt{2}\beta \mp \lambda) X_o^{rs}(2\sqrt{2}\beta \pm \lambda). \quad (71)$$

For their canonical R–NS–R ordering,

$$\rho_{\text{RRe}}^{(\pm)}(w_\beta^+, \nu_d, \chi_{rs}^+) = \rho_{\text{RRe}}^{(\pm)}(\chi_{rs}^+, \nu_d, w_\beta^+) = -P_{rs}^{\text{R},\pm}(d, \beta; c). \quad (72)$$

Other cyclic orderings used in the sphere literature can display an additional $(-1)^{rs/2}$; it is an orientation convention, not a change of the polynomial. The remaining ground-state entries and slot orderings follow from Eqs. (16) and the super-Virasoro Ward identities. Rather than hiding their phases in scalar signs, define at every N–R–R vertex the 2×2 fusion map

$$\mathbf{P}_{rs}^{(v,e)} : \mathcal{G}_{\beta'_{rs}} \longrightarrow \mathcal{G}_{\beta_{rs}}, \quad (73)$$

whose entries are the two polynomials (71) with the phases fixed by the chosen ordered three-point tensor. The four-R sphere formulas of Ref. [3] propagate an NS module and therefore test the distinct polynomial $P_{\text{NS|RR}}^{rs}$ defined below, not this R-internal matrix. The toric formulas of Ref. [5] do test the R-internal kernel. A sphere test of the same matrix would instead use a two-NS/two-R ordering with R exchange and must be derived in the same three-point convention.

5 Sphere four-R calibration with NS exchange

The first Ramond-insertion benchmark is the chiral correlator

$$\langle R_{\beta_4}(\infty) R_{\beta_3}(1) R_{\beta_2}(z) R_{\beta_1}(0) \rangle. \quad (74)$$

The $R \times R$ OPE in this channel propagates an NS module of weight Δ . Consequently this test probes the NS inverse Gram matrix together with the R–R–NS fusion polynomials. It tests the Ramond Ward phases but does not yet test a Ramond internal edge or its G_0 ground-fiber matrix.

Let $\eta_3, \eta_2 \in \{+1, -1\}$ denote the two HJS chiral-form signs. The even and odd blocks are normalized by

$$\begin{aligned} \mathcal{F}_\Delta^e(z) &= z^{\Delta - h_2^{\text{R}} - h_1^{\text{R}}} \left(1 + \sum_{m \geq 1} z^m F_m^e \right), \\ \mathcal{F}_\Delta^o(z) &= z^{\Delta + 1/2 - h_2^{\text{R}} - h_1^{\text{R}}} \left(F_{1/2}^o + \sum_{m \geq 1} z^m F_{m+1/2}^o \right). \end{aligned} \quad (75)$$

For an NS singular vector (r, s) , $r + s$ even, define the fusion polynomial

$$P_{\text{NS|RR}}^{rs} \left[\begin{smallmatrix} \eta\beta_2 \\ \beta_1 \end{smallmatrix} \right] = \prod_{\substack{p=1-r, 3-r, \dots, r-1 \\ q=1-s, 3-s, \dots, s-1 \\ p+q-r-s \in 4\mathbb{Z}+2}} \left(\beta_1 - \eta\beta_2 + \frac{pb + qb^{-1}}{2\sqrt{2}} \right) \times \prod_{\substack{p=1-r, 3-r, \dots, r-1 \\ q=1-s, 3-s, \dots, s-1 \\ p+q-r-s \in 4\mathbb{Z}}} \left(\beta_1 + \eta\beta_2 + \frac{pb + qb^{-1}}{2\sqrt{2}} \right). \quad (76)$$

This polynomial is different from $P_{rs}^{\text{R}, \pm}$ in Eq. (71): the subscript before the vertical bar names the sector of the singular internal module.

At fixed c , write

$$\Delta_{rs} = -\frac{rs-1}{4} + \frac{1-r^2}{8}b^2 + \frac{1-s^2}{8b^2}, \quad K_{rs}^{\eta_3, \eta_2} = A_{rs} P_{\text{NS|RR}}^{rs} \left[\begin{smallmatrix} \eta_3\beta_3 \\ \beta_4 \end{smallmatrix} \right] P_{\text{NS|RR}}^{rs} \left[\begin{smallmatrix} \eta_2\beta_2 \\ \beta_1 \end{smallmatrix} \right], \quad (77)$$

where A_{rs} is Eq. (33) evaluated at the physical b rather than at a fixed- h c -pole. If $\pi \in \{e, o\}$ and $\bar{\pi}$ denotes the opposite parity, the HJS elliptic recursion becomes

$$H_{\Delta}^{\pi}[\eta_3, \eta_2](q) = g^{\pi} + \sum_{\substack{r, s \geq 1 \\ r, s \text{ even}}} \frac{(16q)^{rs/2} K_{rs}^{\eta_3, \eta_2}}{\Delta - \Delta_{rs}} H_{\Delta_{rs} + rs/2}^{\pi}[\eta_3, \eta_2](q) - \sum_{\substack{r, s \geq 1 \\ r, s \text{ odd}}} \frac{(16q)^{rs/2} K_{rs}^{\eta_3, \eta_2}}{\Delta - \Delta_{rs}} H_{\Delta_{rs} + rs/2}^{\bar{\pi}}[-\eta_3, -\eta_2](q), \quad g^e = 1, \quad g^o = 0. \quad (78)$$

The minus sign in the odd-odd Kac sum is fixed, not conventional: the two null-vector three-point phases multiply to $(-e^{i\pi/4})e^{-i\pi/4} = -1$. The elliptic block reconstructed from Eq. (78) is

$$\mathcal{F}_{\Delta}^{\pi}(z) = (16q)^{\Delta - (c-3/2)/24} z^{(c-3/2)/24 - h_1^{\text{R}} - h_2^{\text{R}}} (1-z)^{(c-3/2)/24 - h_2^{\text{R}} - h_3^{\text{R}}} \times \theta_3(q)^{(c-3/2)/2 - 4 \sum_{i=1}^4 h_i^{\text{R}}} H_{\Delta}^{\pi}(q), \quad q = \exp \left[-\pi \frac{K(1-z)}{K(z)} \right]. \quad (79)$$

Two direct Ward-identity coefficients lock the normalization and phase:

$$F_{1/2}^o = -\frac{(\beta_4 - \eta_3\beta_3)(\beta_1 - \eta_2\beta_2)}{2\Delta}, \quad F_1^e = \frac{(\Delta + h_3^{\text{R}} - h_4^{\text{R}})(\Delta + h_2^{\text{R}} - h_1^{\text{R}})}{2\Delta}. \quad (80)$$

In particular, the $(1, 1)$ term of the recursion obeys $[q^{1/2}]H^o/4 = F_{1/2}^o$ because $q = z/16 + O(z^2)$. This is the cheapest sign-sensitive test.

The generic- b implementation in `ramond_sphere_blocks.py` used $c = 27/2 + 10^{-5}$ for the original sign calibration below. The production self-dual wrapper introduced in `self_dual_superconformal_blocks.py` instead combines the resonant terms by Eq. (136). At

$$z = \frac{1}{3} + \frac{3}{5}i, \quad (P_1, P_2, P_3, P_4) = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{3}{5} \right), \quad P_{\text{int}} = \sqrt{\frac{6}{5}}, \quad q \text{ order } 8, \quad (81)$$

the four chiral sign branches are

(η_3, η_2)	$\mathcal{F}^e(z)$	$\mathcal{F}^o(z)$	diagonal e + o
$(+, +)$	$1.1208467965 + 0.1233873315i$	$0.0045330606 + 0.0148096711i$	1.2717618498
$(+, -)$	$1.1215330591 + 0.1232575976i$	$0.0435256242 + 0.0477480711i$	1.2772031963
$(-, +)$	$1.1215225105 + 0.1232595670i$	$0.0203345304 + 0.0242075417i$	1.2740051607
$(-, -)$	$1.1215715280 + 0.1232509064i$	$0.1066695940 + 0.1145408166i$	1.2976114792

Here “diagonal e + o” means $\mathcal{F}^e(z)\mathcal{F}^e(\bar{z}) + \mathcal{F}^o(z)\mathcal{F}^o(\bar{z})$. The direct z -series reproduces Eq. (80), and changing the displayed cutoff from q^6 to q^8 moves each chiral value by less than 10^{-8} at this point.

The last column is deliberately only a diagonal block diagnostic. It is not yet the physical type-0B four-R correlator. In HJS notation the physical R–R–NS vertex already mixes chiral components; for example,

$$R_{\text{NSR}}^+ = \frac{C^{(+)}}{\sqrt{2}} \left(V_{\text{NSRe}}^{(+)} \otimes \bar{V}_{\text{NSRe}}^{(+)} - i V_{\text{NSRo}}^{(+)} \otimes \bar{V}_{\text{NSRo}}^{(+)} \right) + \frac{C^{(-)}}{\sqrt{2}} \left(V_{\text{NSRe}}^{(-)} \otimes \bar{V}_{\text{NSRe}}^{(-)} - i V_{\text{NSRo}}^{(-)} \otimes \bar{V}_{\text{NSRo}}^{(-)} \right), \quad C^{(\pm)} = \frac{C^+ \pm C^-}{2}. \quad (82)$$

Before promoting the diagnostic to a type-0B correlator, these constants and phases must be matched to the BRY-normalized C_{even} and C_{odd} by the three-point convention ledger. The code therefore does not silently identify HJS chiral signs with BRY Ramond-family labels.

6 RRRR and RRNSNS crossing calibration

The physical crossing tests fix the remaining chiral-to-nonchiral ledger. In the delta-normalized positive-momentum BRY basis, define

$$C_+^{\text{RRNS}} = C_{\text{even}}, \quad C_-^{\text{RRNS}} = C_{\text{odd}}. \quad (83)$$

The labels on the left are HJS chiral-form signs, not the two nonchiral type-0B Ramond families. BRY’s two-point normalization is $\langle V_{P_1} V_{P_2} \rangle = \pi \delta(P_1 - P_2)$ in both sectors, so the continuum resolution of the identity carries dP/π . This measure is included explicitly in every factorization formula below. With the ordering $(4, 3, 2, 1) = (\infty, 1, z, 0)$, the four-R correlator is

$$\mathcal{G}_{\text{RRRR}}^{(s)}(z, \bar{z}) = \frac{1}{2} \int_0^\infty \frac{dP}{\pi} \sum_{\eta_3, \eta_2 = \pm 1} C_{\eta_3}^{\text{RRNS}}(P_4, P_3, P) C_{\eta_2}^{\text{RRNS}}(P_2, P_1, P) \times \left(|\mathcal{F}_P^e[\eta_3, \eta_2](z)|^2 + |\mathcal{F}_P^o[\eta_3, \eta_2](z)|^2 \right). \quad (84)$$

Its nontrivial sphere crossing equation is

$$\mathcal{G}_{4321}^{(s)}(z, \bar{z}) = \mathcal{G}_{4123}^{(s)}(1 - z, 1 - \bar{z}). \quad (85)$$

The mixed test compares two different internal sectors. The NS-exchange decomposition of $\langle \phi_4 \phi_3 R_2 R_1 \rangle$ is

$$\mathcal{G}_{\text{NS}}^{(s)}(z, \bar{z}) = \int_0^\infty \frac{dP}{\pi} \sum_{\eta = \pm 1} C_\eta^{\text{RRNS}}(P_2, P_1, P) [C(P_4, P_3, P) |\mathcal{F}_{P, \eta}^e(z)|^2 + \tilde{C}(P_4, P_3, P) |\mathcal{F}_{P, \eta}^o(z)|^2]. \quad (86)$$

Here the phase written as $-i\tilde{C}$ in the HJS component convention has already been absorbed into BRY's real $\tilde{C}$. After crossing, the same correlator is expanded in a generic long-R module,

$$\mathcal{G}_R^{(t)}(1-z, 1-\bar{z}) = \int_0^\infty \frac{dP}{\pi} \sum_{\eta_3, \eta_2 = \pm 1} C_{\eta_3}^{\text{RRNS}}(P_1, P, P_4) C_{\eta_2}^{\text{RRNS}}(P, P_2, P_3) \times |\mathcal{F}_P^{\text{R,e}}[\eta_3, \eta_2](1-z)|^2. \quad (87)$$

Reflection signs and the exchange of C_{even} with C_{odd} on a reflected R leg are canceled by the positive- P BPZ pairing and a dummy relabeling of η_2 . Equations (86) and (87) therefore use only positive momenta. The crossing equation is

$$\mathcal{G}_{\text{NS}}^{(s)}(z, \bar{z}) = \mathcal{G}_R^{(t)}(1-z, 1-\bar{z}). \quad (88)$$

The mixed-R block and correlator modules give the following calibration. For the level-one test, $c = 6$ is used to keep the $(2, 1)$ and $(1, 2)$ poles well separated. For crossing, the type-0B value is approached as $c = 27/2 + \epsilon$.

Test	Cutoff/regulator	Left and right values	Relative error
direct R level one	all four η_3, η_2	Eq. (56) vs recursion	1.2×10^{-16}
RRRR, $z = 0.37$	$q^6, \epsilon = 10^{-5}$	0.2029150374, 0.2029136242	6.96×10^{-6}
RRNSNS, $z = 0.37$	$q^6, \epsilon = 3 \times 10^{-5}$	0.1740316590, 0.1740315492	6.31×10^{-7}
RRNSNS, $z = 0.37$	$q^8, \epsilon = 3 \times 10^{-5}$	0.1740316592, 0.1740315540	6.04×10^{-7}

The mixed residual scales with the $b \rightarrow 1$ displacement and numerical quadrature, while the $q^6 \rightarrow q^8$ change is already below it. The old odd-sublattice implementation of Eq. (69) instead stabilized near a one-percent mismatch. Thus the combined brute-force and crossing tests decisively validate the even-sublattice Ramond residue used here.

Higher recursion order exposes a numerical subtlety at the rational point $c_0 = 27/2$. Separate Kac-residue terms become large as their poles collide, even though their sum is finite. A one-sided displacement that is adequate at low order is consequently ill-conditioned at high order: at recursion order 12, $\epsilon = 3 \times 10^{-5}$ gives a maximum RRNSNS crossing residual 1.07×10^{-4} near $z = 0.95$. We instead use the even regulator extrapolation

$$\mathcal{G}_{\text{sym}}^{(L)}(\epsilon) = \frac{1}{2} \left[\mathcal{G}^{(L)}(c_0 + \epsilon) + \mathcal{G}^{(L)}(c_0 - \epsilon) \right]. \quad (89)$$

It cancels the leading displacement error while keeping the colliding poles far enough apart for double precision. With $\epsilon = 10^{-4}$, $P_{\text{max}} = 5$, 24-point Gauss-Legendre quadrature, and a 37-point grid $0.05 \leq z \leq 0.95$, the maximum relative crossing residual is 1.30×10^{-6} , 1.11×10^{-7} , and 9.31×10^{-8} at recursion orders 8, 10, and 12, respectively. The maximum order-10 $\rightarrow$ 12 drift is 1.88×10^{-8} in the NS channel and 1.76×10^{-8} in the crossed R channel. Thus order 12 is accessible, provided the rational- c collision is regulated symmetrically.

Sphere Ramond milestone. The implementation has now passed two logically independent tests. First, the generic long-R block agrees with a direct Gram-matrix and Ward-identity construction through q^3 , for all four HJS sign branches, with maximum coefficient error 8.77×10^{-13} . Second, the assembled nonchiral correlators satisfy crossing: the documented RRRR benchmark has relative residual 6.96×10^{-6} in its original reduced calibration and 1.32×10^{-8} in the exact- c , order-10 production stress test. The RRNSNS comparison between NS and R exchange has maximum residual 9.31×10^{-8} on the order-12 real- z grid. The mixed test is especially decisive because it changes both the factorization channel and the sector of the exchanged module.

This simultaneously validates the local G_0 convention, Ramond residue normalization, HJS-to-BRY sign dictionary, BRY structure constants and dP/π spectral measure, and the nonchiral channel assembly used on the sphere. It is a genuine project milestone: the generic long-R sphere layer is now calibrated well enough to serve as the numerical oracle for the higher-genus sewing construction. The short module at $P = 0$ and the routing of the open Ramond ground-state fiber around a handle remain separate tasks.

6.1 Legacy fixed-weight c -pole test with Ramond insertions

The sphere Ramond implementation also supplies an independent test of the local central-charge residues. This experiment varied c at fixed Ramond weights. It remains a valid finite- c check of Kac factorization, but it is not the analytic slice used for the Ramond recursion or its regular seed. For an NS internal module, the tested local coefficient at twice-level N was

$$F_N(c) = F_N^{\text{reg}}(c) + \sum_{\substack{r \geq 2, s \geq 1 \\ r+s \in 2\mathbb{Z} \\ rs \leq N}} \frac{\mathcal{R}_{rs}}{c - c_{rs}(h)} F_{N-rs} \left(c_{rs}(h), h + \frac{rs}{2} \right). \quad (90)$$

For RRRR, both adjacent factors in $\mathcal{R}_{rs}$ are the R–R–NS fusion polynomials of Eq. (76). For the NS-exchange channel of NSNSRR, one factor is NS–NS–NS and the other is R–R–NS. The legacy test recomputed the external Ramond parameters at fixed weights,

$$\beta_i^{\text{legacy}}(c_{rs}) = \sqrt{\frac{c_{rs}}{24} - h_i^{\text{R}}}, \quad (91)$$

on the chosen chiral sheet. In the corrected recursion one instead holds

$$\beta_i(c) = \beta_i, \quad h_i^{\text{R}}(c) = \frac{c}{24} - \beta_i^2. \quad (92)$$

The code in `ramond_c_recursive_sphere_blocks.py` tests the residue without using Eq. (90) to generate the comparison block. It first converts the established HJS elliptic h -recursion back to its local z coefficients and then measures

$$\text{Res}_{c=c_{rs}}^{\text{num}} F_N = \frac{\epsilon}{2} [F_N(c_{rs} + \epsilon) - F_N(c_{rs} - \epsilon)]. \quad (93)$$

At $b = 1.27$, with momenta

$$(P_1, P_2, P_3, P_4; P) = (0.31, 0.41, 0.23, 0.37; 0.70)$$

and $\epsilon = 10^{-5}$, the analytic and numerical residues agree as follows:

Block and pole	Sign branches tested	Maximum relative error
RRRR, (3, 1)	$(\eta_3, \eta_2) \in \{\pm 1\}^2$	3.83×10^{-9}
RRRR, (2, 2)	$(\eta_3, \eta_2) \in \{\pm 1\}^2$	1.10×10^{-11}
NSNSRR with NS exchange, (3, 1)	$\eta_2 = \pm 1$	1.36×10^{-9}
NSNSRR with NS exchange, (2, 2)	$\eta_2 = \pm 1$	9.24×10^{-12}

The small relative-error degradation occurs in branches whose residue is close to zero; the absolute agreement remains stable under changing ϵ .

This test fixes the nontrivial local polar kernel: the null-norm factor, R–R–NS fusion polynomials, odd-Kac phases, and the sign flip after an odd null vector. Its fixed-weight Jacobian must be replaced by Eq. (48) in the corrected recursion. The experiment also explains why the fixed-weight route appeared not to close. The usual light-block argument says that L_{-n} , $n > 1$, and G_{-r} , $r > \frac{1}{2}$, are suppressed by their large- c inverse Gram matrix, leaving the global $\mathfrak{osp}(1|2)$ block. This power counting assumes that the external matrix elements remain $O(1)$. At fixed h_i^R , however,

$$G_0^2 = h_i^R - \frac{c}{24}, \quad (94)$$

so one Ramond ground-state combination grows as $\sqrt{c}$. For example, the level-two state $G_{-1/2}G_{-3/2}|h\rangle$ has an $O(c)$ norm but also an $O(c)$ RRNS three-form in the large ground-state channel; its contribution therefore survives.

At fixed β_i , by contrast,

$$G_0^2 = -\beta_i^2 = O(1), \quad (95)$$

so this anomalous $\sqrt{c}$ growth is absent. The failure above is a failure of the fixed- h_R slice, not evidence against a Ramond c -recursion.

The remaining component issue. This statement does not mean that the desired observable has external positive-level descendants. We only want the component in which every external Ramond operator corresponds to a chosen ground vector $|0\rangle$. The difficulty is instead that this scalar component is not closed under the superconformal Ward identities. Put

$$|1\rangle = G_0|0\rangle, \quad \rho_{ab}(I) = \langle R_a(\infty)R_b(1) | I \rangle, \quad a, b \in \{0, 1\}. \quad (96)$$

For an NS internal edge, the required level- N component is

$$F_{0000,N} = \sum_{I,J} \rho_{00}(I) \left(K_N^{\text{NS}} \right)_{IJ}^{-1} \rho_{00}(J). \quad (97)$$

Nevertheless the Ramond OPE

$$G(z)R_0(0) \sim z^{-3/2}R_1(0) + \cdots, \quad R_1 \longleftrightarrow G_0|0\rangle, \quad (98)$$

implies schematically

$$\rho_{00}(G_{-r}I) \in \text{span}\{\rho_{10}(I), \rho_{01}(I)\}. \quad (99)$$

A second supercurrent action can return to the requested component through

$$G_0|1\rangle = G_0^2|0\rangle = \left(h^R - \frac{c}{24} \right) |0\rangle. \quad (100)$$

Thus projecting the external insertions onto $|0\rangle$ removes an unpaired G_0 , but it does not remove even sequences containing G_0^2 . Moreover $G_0|0\rangle$ is not a positive-level descendant: it has the same

L_0 eigenvalue as $|0\rangle$ and is the second vector in the generic Ramond ground multiplet. The closed Ward problem is consequently the four-component system

$$\rho(I) = (\rho_{00}(I), \rho_{01}(I), \rho_{10}(I), \rho_{11}(I))^T, \quad (101)$$

followed only at the end by the projection in Eq. (97).

There is also an important Gram-matrix distinction. In the RRRR and NSNSRR channels discussed here the exchanged module is NS, so K_N^{NS} in Eq. (97) does not invert the metric of the external Ramond ground fibers. If the exchanged module is instead R, its two-dimensional ground metric and all descendant metrics must be included and inverted; factors generated by an internal G_0 can then cancel against that inverse metric. In either case no large- c conclusion should be drawn from an isolated G_0 or Gram-matrix entry: one must form the complete Ward–Gram contraction first.

The correct diagnostic keeps the two Ramond ground indices open. In the normalized HJS basis (w^+, w^-) the two sign-projected ground tensors are

$$B_e^{(\eta)} = \begin{pmatrix} 1 & 0 \\ 0 & \eta \end{pmatrix}, \quad B_o^{(\eta)} = \begin{pmatrix} 0 & 1 \\ i\eta & 0 \end{pmatrix}, \quad \eta = \pm 1, \quad (102)$$

and

$$\mathcal{G}_0(\beta) = i\beta \begin{pmatrix} 0 & e^{i\pi/4} \\ e^{-i\pi/4} & 0 \end{pmatrix}. \quad (103)$$

The Ward recursion immediately gives

$$\left[T_o^{(\eta)}(G_{-1/2}\nu_h) \right]_{++} = e^{-i\pi/4}(\beta_1 - \eta\beta_2), \quad (104)$$

which fixes the phase and sign convention independently of the elliptic recursion.

The implementation in `mixed_ns_ramond_descendant_blocks.py` now constructs the full NS PBW basis, its ordinary- c Gram matrix, the closed NSNS three-form, and the matrix RRNS Ward form. It agrees coefficient by coefficient with the HJS h -recursion for both $\eta = \pm 1$ through local level 13/2. This direct oracle defines the regular term without any h -recursion input:

$$F_N^{\text{reg}}(c) = F_N^{\text{direct}}(c) - \sum_{r,s} \frac{\mathcal{R}_{rs}}{c - c_{rs}(h)} F_{N-rs}^{\text{direct}} \left(c_{rs}(h), h + \frac{rs}{2} \right). \quad (105)$$

The cached implementation is `ramond_c_regular_seed.py`. Supplying it to `ramond_c_recursive_sphere_blocks.py` completes the NSNSRR legacy fixed-weight identity through the tested local level 9/2, agreeing with the independent h -recursion at 3×10^{-12} or better. This is retained as a residue regression only: its F_N^{reg} is not the regular seed of the corrected fixed- β recursion.

The first non-global seed coefficient is already visible at level two. For the benchmark above, at $c = 10^5$,

η	F_4^{reg}	F_4^{global}
+1	0.235669340151	0.235669340362
−1	0.216738667677	0.235669340362

(106)

Thus the ordinary global block is the strict seed in the small ground-state channel, but not in the large one. What remains is to resum the directly determined coefficients into an efficient contracted-Ramond light block; the present Ward/Gram provider is exact but scales combinatorially. For the crossed NSNSRR channel with an internal R module, the internal edge residue is matrix-valued as well.

6.2 Fixed- β contraction of the sphere Ramond channel

The fixed- β boundary problem must be formulated for the actual three-forms entering the sphere block. This corrects an important misidentification. Consider

$$\left\langle V_4^{\text{NS}}(\infty) V_3^{\text{R}}(1) V_2^{\text{R}}(z) V_1^{\text{NS}}(0) \right\rangle \quad (107)$$

factorized through a long Ramond module of momentum β . The two chiral vertices are the ordered HJS NR and RN three-forms,

$$\begin{aligned} \rho_L(I) &= \rho_{NR}^{(\eta_3)} \left(\nu_4, w_{\beta_3}^+, w_{\beta, I}^+ \right), \\ \rho_R(J) &= \rho_{RN}^{(\eta_2)} \left(w_{\beta, J}^+, w_{\beta_2}^+, \nu_1 \right), \end{aligned} \quad (108)$$

not the RR three-form obtained by inserting an NS primary between two Ramond modules. With the even Ramond Gram matrix $B_{R,n}$, the local coefficient is

$$F_n = \sum_{|I|=|J|=n} \rho_L(I) \left(B_{R,n}^{-1} \right)^{IJ} \rho_R(J). \quad (109)$$

These are precisely the definitions underlying the published Ramond elliptic recursion [4].

What becomes of the internal primary. Set

$$D = L_0 - \frac{c}{24}, \quad h_\beta^{\text{R}} = \frac{c}{24} - \beta^2. \quad (110)$$

After fixing the ground metric and identifying the PBW bases at different c , the internal primary has the coefficientwise limit

$$|h_\beta^{\text{R}}, a\rangle \longrightarrow |\beta, a\rangle_\infty = |a\rangle_{\mathcal{R}^{(0)} \hat{\otimes} |0\rangle_{\mathcal{F}}}. \quad (111)$$

For $n > 0$, introduce

$$a_n = \sqrt{\frac{12}{cn^3}} L_n, \quad \psi_n = \sqrt{\frac{3}{cn^2}} G_n, \quad (112)$$

and the corresponding creation modes. On fixed-level states they obey

$$\begin{aligned} [a_n, a_m^\dagger] &= \delta_{nm}, & \{\psi_n, \psi_m^\dagger\} &= \delta_{nm}, \\ D &= -\beta^2 + N, & N &= \sum_{n \geq 1} n \left(a_n^\dagger a_n + \psi_n^\dagger \psi_n \right). \end{aligned} \quad (113)$$

Thus the primary is the oscillator vacuum, but this fact alone does not make either three-form in Eq. (108) trivial.

The surviving G_0 action. Let Γ_β be the odd ground-space matrix satisfying $\Gamma_\beta^2 = -\beta^2$. On the full contracted module,

$$G_0 = \Gamma_\beta + \sum_{n \geq 1} \sqrt{n} \left(a_n^\dagger \psi_n + \psi_n^\dagger a_n \right), \quad G_0^2 = D. \quad (114)$$

In particular,

$$\begin{aligned} G_0 a_n^\dagger |0, a\rangle &= a_n^\dagger \Gamma_\beta |0, a\rangle + \sqrt{n} \psi_n^\dagger |0, a\rangle, \\ G_0 \psi_n^\dagger |0, a\rangle &= -\psi_n^\dagger \Gamma_\beta |0, a\rangle + \sqrt{n} a_n^\dagger |0, a\rangle. \end{aligned} \quad (115)$$

The Ramond ground index must therefore remain open while the two three-forms and inverse Gram matrix are contracted. Replacing G_0 by one number $\pm i\beta$ before this contraction loses the parity mixing.

Why the vertex does not become the identity. In the radial presentation (108), the inserted operator is a Ramond primary. Its weight

$$h_3^R = \frac{c}{24} - \beta_3^2 \quad (116)$$

is $O(c)$ at fixed β_3 . The Virasoro Ward identity gives, for the left three-form,

$$\frac{\rho_L(L_{-n}w_\beta^+)}{\rho_L(w_\beta^+)} = h_\beta^R + nh_3^R - h_4. \quad (117)$$

The right three-form similarly satisfies

$$\frac{\rho_R(L_{-n}w_\beta^+)}{\rho_R(w_\beta^+)} = h_\beta^R + nh_2^R - h_1. \quad (118)$$

Consequently,

$$\begin{aligned} \frac{\rho_L(a_n^\dagger w_\beta^+)}{\rho_L(w_\beta^+)} &= \sqrt{c} \frac{n+1}{4\sqrt{3}n^{3/2}} + O(c^{-1/2}), \\ \frac{\rho_R(a_n^\dagger w_\beta^+)}{\rho_R(w_\beta^+)} &= \sqrt{c} \frac{n+1}{4\sqrt{3}n^{3/2}} + O(c^{-1/2}). \end{aligned} \quad (119)$$

The normalized descendant matrix elements grow as $\sqrt{c}$; they do not vanish. The argument that a fixed-weight NS primary commutes with all contracted oscillators applies to the RR radial vertex, but not to the NR/RN vertices which define the sphere block (107). These presentations are related by a nontrivial change of channel and local coordinates and cannot be interchanged when determining the regular seed.

Level-one certificate including G_0 . In the even basis

$$(L_{-1}w^+, G_{-1}G_0w^+), \quad (120)$$

the exact Gram matrix is

$$B_{R,1} = \begin{pmatrix} 2h_\beta^R & -\frac{3}{2}\beta^2 \\ -\frac{3}{2}\beta^2 & -\beta^2 \left(2h_\beta^R + \frac{c}{4} \right) \end{pmatrix}. \quad (121)$$

The two Ward vectors are

$$\begin{aligned} \boldsymbol{\rho}_L &= \left(h_\beta^R + h_3^R - h_4, -\frac{\beta^2}{2} - \eta_3\beta\beta_3 \right), \\ \boldsymbol{\rho}_R &= \left(h_\beta^R + h_2^R - h_1, -\frac{\beta^2}{2} - \eta_2\beta\beta_2 \right). \end{aligned} \quad (122)$$

The second entries are the explicit G_0 -dependent pieces. Forming the complete Ward–Gram contraction before taking the limit gives

$$F_1^{\text{local}} = \boldsymbol{\rho}_L B_{R,1}^{-1} \boldsymbol{\rho}_R^T = \frac{c}{12} - \beta_2^2 - \beta_3^2 - h_1 - h_4 + O(c^{-1}). \quad (123)$$

This directly disproves the vacuum-only seed for the sphere channel. It also shows why power counting an isolated G_0 matrix element is insufficient: its contribution is determined only after the inverse ground/descendant metric and both Ward vectors are included.

Definition of the elliptic seed. Write

$$q = \exp \left[-\pi \frac{K(1-z)}{K(z)} \right], \quad \Delta_0 = \frac{c - \frac{3}{2}}{24}. \quad (124)$$

The HJS elliptic block is defined, rather than independently postulated, by

$$\mathcal{F}_\beta^{\eta_3, \eta_2}(z) = \mathcal{P}_{\text{ell}}(c, z) H_\beta^{\eta_3, \eta_2}(c; q), \quad H_\beta^{\eta_3, \eta_2}(c; q) = 1 + O(q), \quad (125)$$

where

$$\begin{aligned} \mathcal{P}_{\text{ell}}(c, z) &= (16q)^{h_\beta^R - \Delta_0 - \frac{1}{16}} z^{\Delta_0 - h_1 - h_2^R + \frac{1}{16}} (1-z)^{\Delta_0 - h_2^R - h_3^R} \\ &\times \theta_3(q)^{\frac{c - \frac{3}{2}}{2} - 4(h_1 + h_2^R + h_3^R + h_4) + \frac{1}{2}}. \end{aligned} \quad (126)$$

There is no separate standard object called an ‘‘HJS seed.’’ The fixed β c -recursion requires the large- c boundary of the particular reduced block chosen in (125).

The raw H_β is unbounded coefficientwise. The all-level global-family argument below shows that its leading growth is generated by

$$\mathcal{E}_R(c, q) = \left[\prod_{n=1}^{\infty} \frac{1+q^n}{1-q^n} \right]^{c/6}. \quad (127)$$

We therefore define

$$\hat{H}_\beta(c; q) = \mathcal{E}_R(c, q)^{-1} H_\beta(c; q), \quad S_\beta(q) = \lim_{\substack{c \rightarrow \infty \\ \beta, \beta_2, \beta_3, h_1, h_4 \text{ fixed}}} \hat{H}_\beta(c; q). \quad (128)$$

This definition is tied to the prefactor (126); moving a finite function between the prefactor and H changes S_β . The first two directly determined coefficients are

$$\begin{aligned} S_\beta(q) &= 1 + S_1 q + S_2 q^2 + O(q^3), \\ S_1 &= 8\beta^2 - 8\beta_2^2 - 8\beta_3^2 - 8h_1 - 8h_4 - \frac{1}{2}, \\ S_2 &= 32X^2 - 16\beta^2 + 12\beta_2^2 + 12\beta_3^2 - 4h_1 - 4h_4 - \frac{3}{8}, \\ X &= \beta^2 - \beta_2^2 - \beta_3^2 - h_1 - h_4. \end{aligned} \quad (129)$$

They are independent of (η_3, η_2) through this order, although the G_0 terms in (122) were retained in obtaining the full finite- c block.

All-order derivation. The coherent sources can be removed without guessing mode kernels by decomposing the Ramond module into orthogonal global $\mathfrak{sl}_2$ families before taking the limit. If $\chi_{m,a}$ is an even quasi-primary at level m , let Q_m be its Gram matrix and let

$$K_m(c) = \rho_L(\chi_{m,a})(Q_m^{-1})^{ab} \rho_R(\chi_{m,b}).$$

The exact local block is

$$\mathcal{B}_\beta(c; z) = \sum_{m \geq 0} z^m K_m(c) {}_2F_1 \left(h_\beta^R + m + h_2^R - h_1, h_\beta^R + m + h_3^R - h_4; 2h_\beta^R + 2m; z \right). \quad (130)$$

The R–R presentation makes the large- c power counting transparent. The Möbius map fixes the internal puncture, so its local action there is generated by L_0, L_1 . It therefore acts by a scalar on an L_1 -annihilated quasi-primary and produces none of the coherent lower-level terms seen in a PBW basis. In the R–R presentation the bottom NS intertwiner is the identity on the nonzero contracted oscillators up to $O(c^{-1/2})$. Moving the internal creation modes through it, the corresponding outgoing creation modes kill the external ground-state bra; induction proves that a unit-normalized positive-level quasi-primary has endpoint coupling $O(c^{-1/2})$. Keeping the complete G_0 fibre in this projection gives

$$K_0(c) = 1, \quad K_m(c) = O(c^{-1}) \quad (m \geq 1). \quad (131)$$

This statement can also be proved inductively from the normalized HJS Ward identities: the leading coherent solution is the global surface state, and the common kernel of all contracted annihilators a_n, ψ_n contains no positive-level vector.

Euler’s transformation gives, for every fixed m ,

$$\begin{aligned} {}_2F_1 \left(h_\beta^R + m + h_2^R - h_1, h_\beta^R + m + h_3^R - h_4; 2h_\beta^R + 2m; z \right) \\ = (1-z)^{-c/12+h_1+h_4+\beta_2^2+\beta_3^2} \left[1 + O(c^{-1}) \right]. \end{aligned} \quad (132)$$

Only finitely many m ’s contribute at any fixed power of z . Equations (130)–(132) therefore prove

$$\lim_{c \rightarrow \infty} (1-z)^{c/12} \mathcal{B}_\beta(c; z) = (1-z)^{h_1+h_4+\beta_2^2+\beta_3^2}. \quad (133)$$

Combining this result with (126) and (127) yields the closed elliptic regular term

$$\boxed{S_\beta(q) = \left(\frac{16q}{z} \right)^{\beta^2} (1-z)^{h_1+h_4+\frac{1}{16}\theta_3(q)^{4(h_1+h_4-\beta_2^2-\beta_3^2)+\frac{1}{4}}}.} \quad (134)$$

The direct q, q^2 calculations and the q^3, q^4 HJS comparison are checks of this derivation.

6.3 Direct finite part at the self-dual point

The symmetric displacement is a useful diagnostic, but it is not the production prescription. At $b = 1$ the colliding Kac terms should be added as one meromorphic object before evaluation. On the sphere we implement this directly in the local uniformizer

$$b = e^t, \quad t \longrightarrow 0. \quad (135)$$

For any assembled elliptic coefficient $H_n(b)$, its self-dual value is the constant Laurent coefficient

$$\begin{aligned} H_n^{\text{sd}} &= [t^0] H_n(e^t) = \frac{1}{2\pi i} \oint_{|t|=\rho} \frac{dt}{t} H_n(e^t) \\ &\simeq \frac{1}{N} \sum_{k=0}^{N-1} H_n \left(\exp \left[\rho e^{2\pi i(k+1/2)/N} \right] \right). \end{aligned} \quad (136)$$

The half-step avoids the least-conditioned real and imaginary directions. The trapezoidal rule projects the constant Laurent mode; negative-power pieces from individual resonant residues cancel only after the complete coefficient has been assembled. We evaluate Eq. (136) at two radii, $\rho = 0.04$ and 0.05 , with $N = 24$. Their difference is kept as a collision-resolution error. This is a finite-part projection at $c = 27/2$, not a one-sided $c_0 + \epsilon$ evaluation or an extrapolation in ϵ .

The implementation is `self_dual_superconformal_blocks.py`. It wraps all three sphere recursions presently required by the crossing problem: RRRR with NS exchange, RRNSNS with NS exchange, and RRNSNS with a generic long-R exchange. At order 12 and $z = 0.37$, representative momentum choices give

block	direct self-dual value	
$\mathcal{F}_{\text{RRRR}}^{\text{even}}$	$1.87402925764551 + O(10^{-16}i)$	(137)
$\mathcal{F}_{\text{RRRR}}^{\text{odd}}$	$-0.00504094652288641 + O(10^{-18}i)$	
$\mathcal{F}_{\text{RRNSNS}}^{\text{R-internal}}$	$1.82014451619952 + O(10^{-17}i)$	

The two-radius discrepancy of the most sensitive retained RRRR odd coefficient is below 4×10^{-8} ; the full block is much more stable because the elliptic nome suppresses the highest coefficients. The symmetric- c values agree with the direct finite part through 2.6×10^{-11} for the displayed checks.

The exact- c block has now also been connected to the full physical RRRR spectral integral in `ramond_sphere_correlators.py`. This is the production prescription; the $c_0 \pm \epsilon$ average is retained only as a cheap diagnostic. For

$$(P_1, P_2, P_3, P_4) = (0.20, 0.35, 0.40, 0.55), \quad P_{\max} = 5, \quad (138)$$

the endpoint-and-center grid $z = (0.05, 0.37, 0.95)$ gives

Prescription	q order	P nodes	Maximum residual	Residual at $z = 0.37$
exact $c = 27/2$	8	24	9.86×10^{-7}	1.44×10^{-8}
exact $c = 27/2$	10	24	1.32×10^{-7}	1.47×10^{-8}
exact $c = 27/2$	12	24	1.44×10^{-7}	1.47×10^{-8}
exact $c = 27/2$	10	32	1.32×10^{-8}	2.09×10^{-11}

The apparent order-10 to order-12 plateau at 24 nodes is not a block failure: increasing the spectral quadrature to 32 nodes lowers the central residual by three orders of magnitude and makes the two endpoint residuals comparable. Increasing $P_{\max}$ from 5 to 6 changes the central correlator only at the 10^{-10} level. As a non-special-momentum diagnostic, the distinct tuple $(0.17, 0.28, 0.46, 0.63)$ gives residuals 7.96×10^{-12} at $z = 0.37$ and 4.17×10^{-11} at $z = 0.37 + 0.21i$ in the symmetric-regulator cross-check. The reproducible driver is `stress_rrrr_crossing.py`.

This scan also exposes why the exact prescription matters. At order 12 and $z = 0.05$, the symmetric displaced- c evaluator worsens to 4.08×10^{-7} because of cancellation loss among colliding residues, whereas the direct finite part gives 7.61×10^{-9} with the same 24-node quadrature. Thus the non-monotonic high-order behavior belongs to the regulator, not to RRRR crossing.

7 Genus-one NS/R sewing calibration

The punctured torus is the first topology on which the sector carried by the handle and the operator closing its ground fiber must be specified independently. For one external NS primary ν_λ , the plane-frame definition is

$$\mathcal{F}_{s,h}^\lambda(q) = q^{h-c/24} \sum_n q^n \sum_{|I|=|J|=n} \rho_s(\mathcal{L}_I \nu_h, \nu_\lambda, \mathcal{L}_J \nu_h) (B_{s,h}^{(n)})^{-1}[I, J], \quad s = \text{NS, R}. \quad (139)$$

For an NS handle, $n \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ and the lift $\ell = \pm 1$ of the plumbing parameter acts by

$$q^n \mapsto \ell^{2n} q^n. \quad (140)$$

Thus the same coefficient table produces the NS and $\widetilde{\text{NS}}$ blocks without changing a branch inside the recursion.

The implementation `superconformal_torus_blocks.py` follows the toric recursions of Ref. [5]. For a primary external insertion, the NS elliptic coefficients have the schematic form

$$H_{\Delta,f}^\lambda = \delta_{f0} \sum_{\substack{r+s \in 2\mathbb{Z} \\ rs \leq 2f}} \frac{s_{rs} A_{rs} P_{rs}^{\tau(rs)}(\Delta_{rs} + rs/2, \Delta_\lambda) P_{rs}(\Delta_{rs}, \Delta_\lambda)}{\Delta - \Delta_{rs}} H_{\Delta_{rs}+rs/2, f-rs/2}^\lambda, \quad (141)$$

where $\tau(rs)$ is unstarred for even rs and starred for odd rs . Here $s_{rs} = 1$ for even rs and $s_{rs} = (-1)^{rs} = -1$ for odd rs in the unstarred external block. This gives the direct low-order anchors

$$H_{\Delta,0}^\lambda = 1, \quad H_{\Delta,1/2}^\lambda = -\frac{\Delta_\lambda}{2\Delta}, \quad H_{\Delta,1}^\lambda = \frac{\Delta_\lambda^2}{2\Delta}. \quad (142)$$

The minus sign at half-integer level is the odd- rs sewing sign $s_{rs} = (-1)^{rs}$. It is also fixed without any pole recursion. In the level-1/2 basis $(G_{-1/2}\nu_\Delta)$ and level-1 basis $(L_{-1}\nu_\Delta)$, direct Ward reduction gives

$$B_{1/2} = (2\Delta), \quad \rho_{1/2} = (2\Delta - \Delta_\lambda), \quad B_1 = (2\Delta), \quad \rho_1 = (2\Delta + \Delta_\lambda(\Delta_\lambda - 1)). \quad (143)$$

Hence $F_{1/2} = 1 - \Delta_\lambda/(2\Delta)$ and $F_1 = 1 + \Delta_\lambda(\Delta_\lambda - 1)/(2\Delta)$. Dividing by the NS Verma character $1 + q^{1/2} + q + \dots$ reproduces Eq. (142). This agrees with the independent descendant ledger of Ref. [6]; the isolated positive sign for $H_{1/2}$ printed in Appendix C of Ref. [5] is incompatible with its odd- rs sign and with direct sewing.

For a generic long-R handle, levels are integral and the two HJS sign components satisfy

$$H_{\beta,\pm,f}^\lambda = \delta_{f0} \sum_{\substack{r+s \in 2\mathbb{Z}+1 \\ rs \leq 2f}} \frac{A_{rs}^{\text{R},(h)} P_{rs}^{\text{R}}(\pm\beta'_{rs}, \Delta_\lambda) P_{rs}^{\text{R}}(\pm\beta_{rs}, \Delta_\lambda)}{\Delta_\beta - \Delta_{rs}^{\text{R}}} H_{\beta'_{rs},\pm,f-rs/2}^\lambda. \quad (144)$$

The code uses $A_{rs}^{\text{R},(h)} = -2\beta_{rs} A_{rs}^{\text{R},(\beta)}$, so the sphere beta-pole residue is not accidentally inserted into the torus weight-pole formula.

There is an equally direct R-sector test. Write $\kappa^2 = \Delta - c/24$ and use the even level-one basis $(L_{-1}w^+, G_{-1}G_0w^+)$. Super-Virasoro normal ordering and the RR Ward identities give

$$B_1^{\text{R}} = \begin{pmatrix} 2\Delta & \frac{3}{2}\kappa^2 \\ \frac{3}{2}\kappa^2 & \kappa^2(2\Delta + c/4) \end{pmatrix}, \quad (145)$$

and, for the HJS sign $s = \pm 1$,

$$\rho_{1,s}^{\text{R}} = \begin{pmatrix} 2\Delta + \Delta_\lambda(\Delta_\lambda - 1) & \kappa^2[\frac{3}{2} + (\Delta_\lambda - 1)(1 - s)] \\ \kappa^2[\frac{3}{2} + (\Delta_\lambda - 1)(1 - s)] & s\kappa^2(2\Delta + c/4 - 2\Delta_\lambda) \end{pmatrix}. \quad (146)$$

The brute-force coefficient is $F_{1,s}^{\text{R}} = \text{Tr}[(B_1^{\text{R}})^{-1}\rho_{1,s}^{\text{R}}]$. It agrees algebraically and numerically with both level-one R recursions.

Crucially, the R handle is not represented only by the scalar sign label. In the unnormalized ground basis (w^+, G_0w^+) , the code stores

$$B_0 = \begin{pmatrix} 1 & 0 \\ 0 & \kappa^2 \end{pmatrix}, \quad G_0 = \begin{pmatrix} 0 & \kappa^2 \\ 1 & 0 \end{pmatrix}, \quad (-1)^F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \kappa^2 = h - \frac{c}{24}. \quad (147)$$

The normalized even HJS ground vertices are $V_{\pm} = \text{diag}(1, \pm\kappa^2)$. Closing the handle only after forming this matrix gives

$$\begin{array}{c|cc} & \text{Tr}(B_0^{-1}V_{\pm}) & \text{Tr}(B_0^{-1}(-1)^F V_{\pm}) \\ \hline V_+ & 2 & 0 \\ V_- & 0 & 2. \end{array} \quad (148)$$

An insertion of G_0 has zero trace for the even external primary. The odd external component and the odd-supermodulus measure must be treated before interpreting that zero as a physical odd-spin contribution.

The production wrappers are

`SelfDualNSTorusOnePointBlock`
`SelfDualRamondTorusOnePointBlock.`

They take the finite part of every assembled coefficient at $b = 1$. For

$$(P_{\text{int}}^{\text{NS}}, P_{\text{int}}^{\text{R}}, P_{\text{ext}}) = (0.61, 0.60, 0.33), \quad (149)$$

the first exact Type-0B coefficient tables are

$$\begin{aligned} H_{\text{NS}}(q) &= 1 - 0.404088623278187 q^{1/2} + 0.224046937176591 q \\ &\quad - 0.446008959609152 q^{3/2} + 0.123538673719732 q^2 + \dots, \\ H_{\text{R},+}(q) &= 1 - 0.417977945105034 q + 0.0252790989890053 q^2 - 0.364113291977004 q^3 + \dots. \end{aligned} \quad (150)$$

The two-radius finite-part disagreement through the displayed orders is below 1.3×10^{-13} . Eight tests in `test_superconformal_torus_blocks.py` verify the NS character and lift, the direct matrices in Eqs. (143) and (146), both HJS Ramond level-one formulas, and the exact- c projections. The recursive ledger is `evaluate_superconformal_torus_block.py`; its independent leading-order comparison is `compare_superconformal_torus_leading.py`.

7.1 Two primary insertions: the NS necklace channel

The first multipuncture test uses two bottom-component NS primaries of weights d_1, d_2 . Cut the torus into two pairs of pants, put NS modules of weights h_1, h_2 on the two internal edges, and denote the corresponding plumbing parameters by q_1, q_2 . In the annulus frame of Ref. [6], the insertions are at $1, q_1$ and the torus nome is

$$q = q_1 q_2. \quad (151)$$

The chiral necklace block is

$$\begin{aligned} \mathcal{F}_{\text{NSNS}}^{(2)}(q_1, q_2) &= q_1^{h_1-c/24} q_2^{h_2-c/24} \sum_{I_a, J_a} q_1^{|I_1|} q_2^{|I_2|} (B_{h_1}^{-1})^{I_1 J_1} (B_{h_2}^{-1})^{I_2 J_2} \\ &\quad \times \rho(\mathcal{L}_{J_1} \nu_{h_1}, \nu_{d_1}, \mathcal{L}_{I_2} \nu_{h_2}) \rho(\mathcal{L}_{J_2} \nu_{h_2}, \nu_{d_2}, \mathcal{L}_{I_1} \nu_{h_1}). \end{aligned} \quad (152)$$

The lift $\ell_a = \pm 1$ on edge a multiplies a term at level n_a by $\ell_a^{2n_a}$.

Direct Gram/Ward sewing gives the complete leading global ledger

$$\begin{aligned} \mathcal{F}_{\text{NSNS}}^{(2)} &= q_1^{h_1-c/24} q_2^{h_2-c/24} \left[1 + A_{10} q_1 + A_{01} q_2 + \ell_1 \ell_2 A_{\frac{1}{2}\frac{1}{2}} (q_1 q_2)^{1/2} + \dots \right], \\ A_{10} &= \frac{(h_1 + d_1 - h_2)(h_1 + d_2 - h_2)}{2h_1}, \quad A_{01} = \frac{(h_2 + d_1 - h_1)(h_2 + d_2 - h_1)}{2h_2}, \\ A_{\frac{1}{2}\frac{1}{2}} &= \frac{(h_1 + h_2 - d_1)(h_1 + h_2 - d_2)}{4h_1 h_2}. \end{aligned} \quad (153)$$

There is no isolated $q_1^{1/2}$ or $q_2^{1/2}$ term: one odd internal state cannot couple at both ends to even external primaries. The mixed term uses

$$\rho(G_{-1/2}\nu_{h_i}, \nu_d, G_{-1/2}\nu_{h_j}) = h_i + h_j - d. \quad (154)$$

As a stringent degeneration check, set $d_1 = 0$ and $h_1 = h_2 = h$. Then $A_{10} = A_{01} = 0$ and $A_{\frac{1}{2}\frac{1}{2}} = 1 - d_2/(2h)$, exactly the raw torus one-point coefficient $F_{1/2}$ with nome $q = q_1 q_2$.

For the full block it is convenient to recurse in $h = h_1$ at fixed $a = h_2 - h_1$. Write $m_i = 2n_i$ for twice the level and retain a routing label $\sigma \in \mathbb{Z}_2$. Its regular part is

$$F_{m_1, m_2}^{(\sigma)} \Big|_{\text{reg}} = \delta_{\sigma, 0} \delta_{m_1, m_2} d_{\text{NS}}(m_1), \quad (155)$$

where $d_{\text{NS}}(m)$ is the NS Verma-character coefficient. A pole (r, s) , with $r + s$ even and $p = rs$, on the first or second edge calls the same coefficient at

	m'_1	m'_2	(h', a', σ')
e_1	$m_1 - p$	m_2	$(\Delta_{rs} + p/2, a - p/2, \sigma + p \bmod 2)$
e_2	m_1	$m_2 - p$	$(\Delta_{rs} - a, a + p/2, \sigma + p \bmod 2)$

(156)

The two denominators are $h - \Delta_{rs}$ and $h + a - \Delta_{rs}$. Each residue is A_{rs} times the two endpoint fusion polynomials, both chosen from the unstarred or starred family specified by σ . The routing flip for odd p is essential: without it the recursion produces spurious isolated $q_i^{1/2}$ terms.

At $b = 1$ the code takes the constant Laurent coefficient of each completely assembled bivariate coefficient on $b = e^t$. When $a = 0$, recursive pole divisors also coincide, so an inner Cauchy projection in a is performed before the b projection. This ordering reproduces the one-point torus recursion when one puncture is the identity.

For delta-normalized Type-0B operators, the corresponding sector-resolved nonchiral NS contribution is

$$\begin{aligned} \langle V_{P_1}(1) V_{P_2}(q_1) \rangle_{\tau, \text{NS}} &= \int_0^\infty \frac{dP_a}{\pi} \int_0^\infty \frac{dP_b}{\pi} C(P_a, P_1, P_b) C(P_b, P_2, P_a) \\ &\times \left| \mathcal{F}_{\text{NSNS}}^{(2)}(P_a, P_b; P_1, P_2; q_1, q_2) \right|^2. \end{aligned} \quad (157)$$

The code `superconformal_torus_two_point.py` implements Eqs. (153)–(156). The companion evaluator `evaluate_superconformal_torus_two_point.py` prints either the direct leading ledger or the finite-part h-recursive coefficient table. For

$$(P_a, P_b; P_1, P_2) = (0.61, 0.74; 0.33, 0.41), \quad (q_1, q_2) = (0.08, 0.05),$$

it gives

$$(A_{10}, A_{01}, A_{\frac{1}{2}\frac{1}{2}}) = (0.168809277749435, 0.278773559059188, 0.373422898519925)$$

and a leading descendant series 1.05106075796861. Equation (157) is numerically available through `super_liouville_torus_two_point.py` and `evaluate_super_liouville_torus_two_point.py`; the level cutoffs on the two edges, momentum cutoff, and spectral quadrature order are independently adjustable. As a normalization anchor, the direct leading block at the displayed external momenta and plumbing parameters, $P_{\text{max}} = 3.5$, and 12-point quadrature gives

$$\langle V_{0.33}(1) V_{0.41}(0.08) \rangle_{\tau, \text{NS}}^{\text{leading}} = 0.00644006440201.$$

This number has a stable spectral quadrature but carries the truncation error of the leading block. The recursive evaluator replaces it by a systematically convergent bivariate level truncation; a claimed correlator value must still include separate level, P_{max} , quadrature, and finite-part convergence checks.

7.1.1 Ramond necklace: ground-fiber layer

For Ramond sewing both internal levels are integral, but level zero is already non-scalar. On edge $a = 1, 2$ retain the unnormalized long-module fiber

$$E_a = \text{span}\{e_{a,+}, e_{a,-}\}, \quad e_{a,+} = w_a^+, \quad e_{a,-} = G_0 w_a^+, \quad B_a^{(0)} = \begin{pmatrix} 1 & 0 \\ 0 & \kappa_a^2 \end{pmatrix}, \quad \kappa_a^2 = h_a - \frac{c}{24}.$$

At the Type-0B continuum point one may take $\kappa_a = P_a/\sqrt{2} > 0$. Suchanek's normalized even RRNS forms are $\rho_{\text{RR},e}^{(\sigma)} = \rho_{\text{RR},e}^{++} + \sigma \rho_{\text{RR},e}^{--}$ with $\sigma = \pm 1$ [4]. Transferring their ground matrix $\text{diag}(1, \sigma)$ to the E_a basis gives the vertex from edge b to edge a ,

$$V_d^{(\sigma)}(a, b)|_{0,0} = \begin{pmatrix} 1 & 0 \\ 0 & \sigma \kappa_a \kappa_b \end{pmatrix}. \quad (158)$$

The sign σ is the HJS chiral-form label, not the nonchiral Type-0B $V^{\text{R},\pm}$ family label.

Let S_a be the chosen identity, parity, G_0 , or parity- G_0 insertion on plumbing edge a . The level-zero necklace coefficient must be formed before making any scalar sign projection:

$$A_{00}^{\sigma_1 \sigma_2}[S_1, S_2] = \text{Tr}_{E_1} \left[(B_1^{(0)})^{-1} S_1 V_{d_1}^{(\sigma_1)}(1, 2) (B_2^{(0)})^{-1} S_2 V_{d_2}^{(\sigma_2)}(2, 1) \right]. \quad (159)$$

In particular,

$$A_{00}^{\sigma_1 \sigma_2}[\mathbf{1}, \mathbf{1}] = 1 + \sigma_1 \sigma_2, \quad A_{00}^{\sigma_1 \sigma_2}[(-1)^F, \mathbf{1}] = 1 - \sigma_1 \sigma_2. \quad (160)$$

Thus the identity cycle keeps equal HJS signs while one parity insertion keeps opposite signs. If $d_1 = 0$, $h_1 = h_2 = h$, and $\sigma_1 = +1$, then $V_0^{(+)} = B^{(0)}$ and Eq. (159) reduces exactly to the Ramond torus one-point ground trace.

The R recursion is most naturally written in β_i , where $h_i = c/24 - \beta_i^2$. For $r + s$ odd, $p = rs$ is even and the null state occurs at integer level $p/2$. If edge e is active and $\bar{e}$ is the adjacent edge, define

$$\Pi_{e,rs}^{\sigma_1 \sigma_2} = P_{\text{R|RNS}}^{rs}(\beta_{\bar{e}}, d_1; \sigma_1) P_{\text{R|RNS}}^{rs}(\beta_{\bar{e}}, d_2; \sigma_2).$$

The normalized coefficient recursion contains the paired residues

$$\begin{aligned} F_{\vec{n}}(\beta_1, \beta_2; \sigma_1, \sigma_2) &= G_{\vec{n}}(\beta_1, \beta_2; \sigma_1, \sigma_2) \\ &+ \sum_{e=1}^2 \sum_{\substack{r,s \geq 1 \\ r+s \text{ odd} \\ rs/2 \leq n_e}} A_{rs}^{\text{R}} \left[\frac{\Pi_{e,rs}^{\sigma_1 \sigma_2}}{\beta_e - \beta_{rs}} F_{\vec{n} - \frac{rs}{2} \vec{e}}(\beta_e \rightarrow \beta'_{rs}; \sigma_1, \sigma_2) \right. \\ &\quad \left. - \frac{\Pi_{e,rs}^{-\sigma_1, -\sigma_2}}{\beta_e + \beta_{rs}} F_{\vec{n} - \frac{rs}{2} \vec{e}}(\beta_e \rightarrow \beta'_{rs}; -\sigma_1, -\sigma_2) \right]. \end{aligned} \quad (161)$$

Thus the negative beta pole flips both adjacent HJS signs and preserves the ground-trace selection rule. Direct symmetric residue measurements at the $(1, 2)$ and $(2, 1)$ poles reproduce Eq. (161).

Unlike the scalar sphere R-exchange elliptic block, the closed necklace does not have $G_{\vec{n}} = \delta_{\vec{n},0}$. The G_0 ground matrices leave a pole-free function of the adjacent beta. The current implementation extracts this regular part from independent two-leg Ward/Gram sewing. This is validated through level one on each edge; deriving its contracted all-level R light block is the remaining step to an efficient high-order recursion.

The resulting first term of the chiral block is

$$\mathcal{F}_{\text{RR}}^{(2)}(q_1, q_2) = q_1^{h_1-c/24} q_2^{h_2-c/24} [A_{00}^{\sigma_1\sigma_2}[S_1, S_2] + A_{10}q_1 + A_{01}q_2 + \cdots]. \quad (162)$$

There are no half-integer powers on either R edge. At level one the full long-module space contains both parity blocks,

$$\mathcal{H}_{\text{R},1}^{(+)} = \text{span}\{L_{-1}w^+, G_{-1}G_0w^+\}, \quad \mathcal{H}_{\text{R},1}^{(-)} = \text{span}\{L_{-1}G_0w^+, G_{-1}w^+\}.$$

Consequently A_{10} and A_{01} require both level-one Gram blocks and the corresponding unequal-weight RRNS Ward matrices. These matrices now supply the regular seed in Eq. (161); the two scalar HJS sign blocks alone would not contain sufficient data.

The generic and self-dual R two-point recursion classes in `superconformal_torus_two_point.py` implement Eqs. (158)–(161), including coefficient-wise finite parts at $b = 1$. The evaluator `evaluate_ramond_torus_two_point.py` uses by default

$$(P_a, P_b; P_1, P_2) = (0.61, 0.74; 0.33, 0.41), \quad (q_1, q_2) = (0.08, 0.05).$$

For identity insertions and $(\sigma_1, \sigma_2) = (+, +)$ it gives

$$(A_{00}, A_{01}, A_{10}, A_{11}) = (2, 0.535870853296445, 0.348609505058044, 2.53432849144586)$$

and the normalized level-one chiral block is 0.284151531566566. The regression suite checks both beta-pole signs, all four nonzero sign/cycle sectors, cyclic exchange, and identity-puncture reduction to the R one-point block.

The R-handle contribution to the nonchiral Type-0B two-point function is assembled as

$$\sum_{\substack{\sigma_1, \sigma_2 = \pm \\ \sigma_1 \sigma_2 = (-1)^{f_1 + f_2}}} \int_0^\infty \frac{dP_a}{\pi} \frac{dP_b}{\pi} 2 C_{\sigma_1}(P_a, P_b; P_1) C_{\sigma_2}(P_b, P_a; P_2) \left| \hat{\mathcal{F}}_{\text{RR}}^{(2), \sigma_1 \sigma_2} \right|^2, \quad (163)$$

where f_a records a parity insertion and the hat denotes normalization to ground coefficient one. The factor two is the closed long-R ground multiplicity and is inserted outside the absolute square. `Type0BRamondTorusTwoPointBetaRecursionCorrelator` implements Eq. (163) at the validated level-one cutoff.

7.2 One-point modular frame change

A fixed- P torus block is not modular covariant by itself: the modular kernel mixes internal representations. The correct test is the assembled spectral integral. The ordinary NS trace, with no $(-1)^F$ insertion, is fixed as a spin structure by S . For an external NS primary of chiral weight d , define

$$\mathcal{G}_{\text{NS}}(\tau) = \int_0^\infty \frac{dP}{\pi} C(P, P_{\text{ext}}, P) \left| q^{h(P)-c/24} F_P^{\text{NS}}(q) \right|^2, \quad q = e^{2\pi i \tau}. \quad (164)$$

Changing the plumbing frame by

$$\tau \mapsto -\frac{1}{\tau}, \quad q \mapsto \tilde{q} = \exp\left(-\frac{2\pi i}{\tau}\right) \quad (165)$$

must give

$$\mathcal{G}_{\text{NS}}(-1/\tau) = |\tau|^{2d} \mathcal{G}_{\text{NS}}(\tau). \quad (166)$$

The Ramond-sewn companion is not obstructed by missing pants data: the RRNS coefficients C_{even} and C_{odd} , together with their HJS chiral-sign dictionary, were already fixed by the sphere crossing tests. For a bottom-component external NS primary, the cyclic contraction can in fact be reduced to the scalar HJS blocks. Their even/odd Ramond subblocks obey

$$\mathcal{F}_{\beta,e}^{\lambda(\pm)}(q) = \pm \mathcal{F}_{\beta,o}^{\lambda(\pm)}(q). \quad (167)$$

Consequently the identity cycle kills the HJS minus branch and doubles the plus branch in each chirality. HJS write the surviving nonchiral coefficient as $4C_{\text{HJS}}^{(+)}$. In the present BRY family convention,

$$C_{\text{HJS}}^{(+)} = \frac{C_+^{\text{RRNS}} + C_-^{\text{RRNS}}}{2} = \frac{C_{\text{even}}}{2}.$$

The completely normalized spectral formulas are therefore

$$\mathcal{G}_{\widetilde{\text{NS}}}(\tau) = \int_0^\infty \frac{dP}{\pi} C(P, P_{\text{ext}}, P) \left| q^{h_{\text{NS}}(P)-c/24} F_P^{\text{NS}}(q; -\epsilon_\tau) \right|^2, \quad (168)$$

$$\mathcal{G}_{\text{R}}(\tau) = 2 \int_0^\infty \frac{dP}{\pi} C_{\text{even}}(P, P; P_{\text{ext}}) \left| q^{h_{\text{R}}(P)-c/24} F_{P,e}^{\text{R},+}(q) \right|^2. \quad (169)$$

The factor two in Eq. (169) is thus fixed by the cyclic contraction and the HJS-to-BRY coefficient map; it is not a normalization fitted to modularity. This reduction is special to a one-point trace with an even external primary. Multipuncture R sewing and higher-genus plumbing must still retain the open Ramond ground fibers before the cycle operators are chosen.

The second even-spin modular orbit is

$$\widetilde{\text{NS}} \xleftrightarrow{S} \text{R}. \quad (170)$$

In the direct-to-transformed orientation used below,

$$\mathcal{G}_{\text{R}}(-1/\tau) = |\tau|^{2d} \mathcal{G}_{\widetilde{\text{NS}}}(\tau). \quad (171)$$

Here the reduced coordinate q does not by itself specify the NS plumbing lift: τ and $\tau + 1$ give the same q but opposite values of $q^{1/2}$. In each modular frame the implementation therefore uses

$$q_{\text{NS}}^{1/2} = e^{\pi i \tau} = \epsilon_\tau \sqrt[q]{q}, \quad \epsilon_\tau \in \{+1, -1\}, \quad (172)$$

and determines ϵ_τ and $\epsilon_{-1/\tau}$ independently. This preserves the chosen spin structure even when either modular parameter lies outside the principal strip. The Casimir factor $q^{h-c/24}$ and the NS Verma character restored in F_P^{NS} are essential; testing only the elliptic remainder H_P would not check Eq. (166).

The implementation `super_liouville_torus_one_point.py` performs the BRY-normalized dP/π integral using the exact- $c = 27/2$ finite-part block, and `check_super_liouville_torus_modular.py` compares the two frames. At

$$\tau = 0.2 + 0.9i, \quad P_{\text{ext}} = 0.33, \quad d = 0.55445,$$

the order-12 calculation means maximum twice-level 12, hence every NS term through q^6 . It gives

$$\begin{aligned} \mathcal{G}_{\text{NS}}(q) &= 0.0546233246078005, \\ \mathcal{G}_{\text{NS}}(\tilde{q}) &= 0.0499165375505112, \\ \frac{\mathcal{G}_{\text{NS}}(\tilde{q})}{\mathcal{G}_{\text{NS}}(q)} &= 0.913831919036706, \quad |\tau|^{2d} = 0.913831919036708. \end{aligned} \quad (173)$$

At fixed 48-point spectral quadrature, the recursion convergence and modular residual are

maximum twice-level	6	8	10	12
relative modular mismatch	6.87×10^{-9}	1.16×10^{-10}	6.40×10^{-13}	1.55×10^{-15}

with $P_{\max} = 4.5$. The order-10 $\rightarrow$ 12 displacement of $\mathcal{G}_{\text{NS}}(q)$ is 6.40×10^{-13} relative, independently confirming that the recursion rather than only the modular ratio has converged. The two finite-part radii differ by less than 1.9×10^{-13} in absolute coefficient value through this order on the sampled momentum range. This is a two-radius stability diagnostic, not a rigorous absolute-error bound; changing the contour sample count moves some level-12 coefficients by up to 2.85×10^{-13} without a visible change in the integrated benchmark. The command `check_super_liouville_torus_modular.py` with `-show-convergence` reproduces the shared-quadrature ledger. This tests both the exact- c recursion and the change of plumbing frame.

The second orbit is implemented by the same module and exposed through `check_super_liouville_torus_spin_modular.py`. The Cannon production run used 96 Gauss–Legendre nodes, $P_{\max} = 5.5$, 50-decimal structure constants, and 40 finite-part contour samples. At the principal benchmark,

$$\begin{aligned}\widetilde{\mathcal{G}}_{\text{NS}}(0.2 + 0.9i) &= 0.049350291217836635, \\ \mathcal{G}_{\text{R}}(-1/(0.2 + 0.9i)) &= 0.04509787132861643,\end{aligned}\tag{174}$$

and the relative error in Eq. (171) is 8.88×10^{-15} . At the harder nonprincipal point $\tau = 0.45 + 0.65i$, the base and $\widetilde{\text{NS}}$ lift signs are $(+1, -1)$ and the convergence is

maximum twice-level	6	8	10	12
relative modular mismatch	5.10×10^{-6}	1.62×10^{-7}	4.23×10^{-9}	1.01×10^{-10}

The order-12 values there are

$$\widetilde{\mathcal{G}}_{\text{NS}} = 0.059304257619173906, \quad \mathcal{G}_{\text{R}} = 0.04569951060828269.$$

The 64-to-96 node comparison changes the correlators by at most 4.62×10^{-15} , changing $P_{\max}$ from 4.5 to 5.5 changes them by at most 9.84×10^{-16} , and the production two-radius diagnostic is 4.89×10^{-13} . All configured numerical and modularity gates pass. The retrieved fingerprinted record is `../DataSet/results/type0b_torus_modular_cluster/cannon_ns_tilde_r_20260723_v1/`.

To display the second-orbit equality as a function rather than only at the two anchor points, Cannon array job 34617773 and reducer 34617786 scanned

$$\tau = 0.2 + iy, \quad y = 0.45, 0.50, \dots, 1.40$$

at maximum twice-level 12. Figure 1 overlays the direct $\widetilde{\text{NS}}$ result with the modular-weight-corrected Ramond result. The residual is 9.00×10^{-8} at the most demanding $y = 0.45$ endpoint, falls below 4.0×10^{-12} by $y = 0.70$, and is 1.15×10^{-12} at $y = 1.40$. The 64-to-96 node comparison changes the assembled correlators by at most 8.93×10^{-15} , so the visible endpoint loss is recursion truncation rather than spectral quadrature. All configured modularity, quadrature, and finite-part gates pass. The summary, submission record, and plotting data are stored under `../DataSet/results/type0b_torus_modular_cluster/cannon_ns_tilde_r_qscan_20260723_v1/`.

For the accuracy run, the spectral integral is parallelized over Gauss–Legendre momentum nodes by `super_liouville_torus_modular_cluster.py`. One worker evaluates the BRY structure constant and the exact- c finite-part coefficients at one node, then reuses them for every requested recursion cutoff and modular frame. Array tasks write disjoint JSONL shards, and the dependent reduction job validates the configuration hash, checks that every node occurs exactly once, sorts

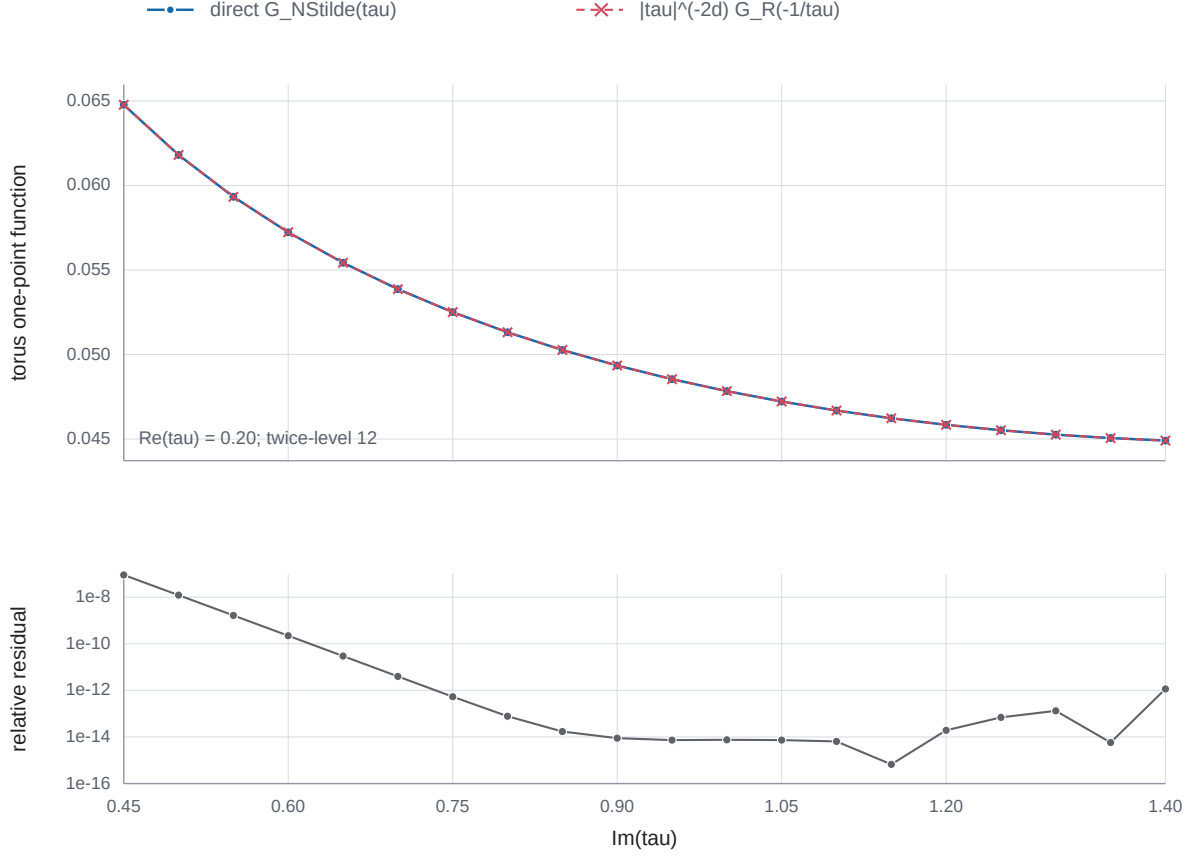


Figure 1: Two-channel Cannon scan of the $\widetilde{\text{NS}} \xleftrightarrow{S} \text{R}$ torus one-point modularity relation at $P_{\text{ext}} = 0.33$, $\text{Re } \tau = 0.20$, and maximum twice-level 12. The upper panel overlays the direct $\mathcal{G}_{\widetilde{\text{NS}}}(\tau)$ evaluation (solid circles) with $|\tau|^{-2d} \mathcal{G}_{\text{R}}(-1/\tau)$ (dashed crosses). The lower panel resolves their relative residual on a logarithmic scale.

by quadrature index, and performs compensated real and imaginary sums. Artifact schema 2 additionally requires an exact implementation fingerprint: SHA-256 hashes of the numerical source closure together with the Python, NumPy, and `mpmath` versions are recorded in every shard, checked before cache reuse and reduction, and copied into the final summary. Thus the parallel result is deterministic and involves no shared-file append or silent reuse across code changes.

The default production ledger has 352 node jobs in 16 shards. It compares 64 and 96 spectral nodes, $P_{\max} = 4.5$ and 5.5 , and 32 and 40 finite-part contour samples at 50-decimal structure-constant precision. For both $\tau = 0.2 + 0.9i$ and the nonprincipal-lift point $\tau = 0.45 + 0.65i$, it evaluates maximum twice-levels 6, 8, 10, 12 from the same prepared coefficients. A complete eight-process simulation gives

$$\delta_{\text{mod}}^{\text{principal}} = 1.55 \times 10^{-15}, \quad \delta_{\text{mod}}^{\text{nonprincipal}} = 1.13 \times 10^{-10},$$

while the $64 \rightarrow 96$ quadrature and $P_{\max} : 4.5 \rightarrow 5.5$ changes are below 1.5×10^{-15} . The largest production two-radius diagnostic is 7.75×10^{-14} . The versioned accuracy configuration and Slurm launch instructions are in `../../../../Code/config/type0b_torus_modular_cluster.json` and `cluster/README_type0b_torus_modular.md`.

The Cannon production run, array job 34584457 followed by reducer 34584458, completed all 352 nodes with zero exit status. Its order-12 production values are

$$\begin{aligned} \mathcal{G}_{\text{NS}}(0.2 + 0.9i) &= 0.05462332460780035, \\ \mathcal{G}_{\text{NS}}(-1/(0.2 + 0.9i)) &= 0.049916537550511134, \end{aligned} \tag{175}$$

with relative modular residual 1.44329×10^{-15} . At the nonprincipal-lift point $\tau = 0.45 + 0.65i$, the independently reconstructed lifts are $(+1, -1)$ and the two frame values are 0.06179554626182613 and 0.04761928293035410, with residual 1.13284×10^{-10} . The summary and submission record are stored under `../../../../DataSet/results/type0b_torus_modular_cluster/cannon_q12_20260723_v1/`. This historical run predates artifact-schema-2 implementation fingerprinting: its configuration digest is verified, but the stored numbers are not cryptographically bound to a source and environment manifest. A fresh run with the current driver is required for that stronger certification.

The isolated machine-precision agreement at the first benchmark should not be mistaken for a generic exact finite-order identity. Cannon array job 34586419 and reducer 34586420 scan 19 modular parameters. For the family $\tau = 0.2 + iy$, the order-12 residual changes as

y	0.45	0.55	0.65	0.75	0.85
δ_{mod}	1.98×10^{-7}	3.00×10^{-9}	4.66×10^{-11}	7.37×10^{-13}	1.24×10^{-14}

With the explicit floor exclusion $\delta_{\text{mod}} > 10^{-15}$, seven radial points enter the reproducible log-log fit

$$\delta_{\text{mod}} = 16.0691 q_{\text{max}}^{6.461873097...}, \quad q_{\text{max}} = \max(|q|, |\tilde{q}|).$$

The fitted exponent is consistent with the expected scaling from the first omitted NS half-level $q^{6.5}$ after truncation through q^6 . The plotting script records the floor, included-point ledger, normalization, and exponent in `torus_modularity_q_scan.fit.json`. At larger y , $|\tilde{q}|$ becomes the larger nome and the residual rises again, producing the modularly symmetric valley in the scan.

The second family, $\tau = 0.45 + iy$, crosses the transformed-lift boundary

$$\text{Re}(-1/\tau) = -\frac{1}{2} \iff y = \sqrt{0.6975} = 0.83516 \dots$$

The reconstructed lift changes from $(+1, -1)$ to $(+1, +1)$ without any discontinuity in the correlator or residual. The 64-to-96 node comparison changes the assembled correlators by at most

3.19×10^{-15} , so the observed pattern is recursion truncation rather than spectral quadrature. The historical q -scan reduction gated quadrature and finite-part stability, but not the modular residual itself. The current scan configuration adds order-12 modular-residual gates at the representative principal point $\tau = 0.20 + 0.85i$ and nonprincipal point $\tau = 0.45 + 0.65i$; these become certified checks on the next artifact-schema-2 run.

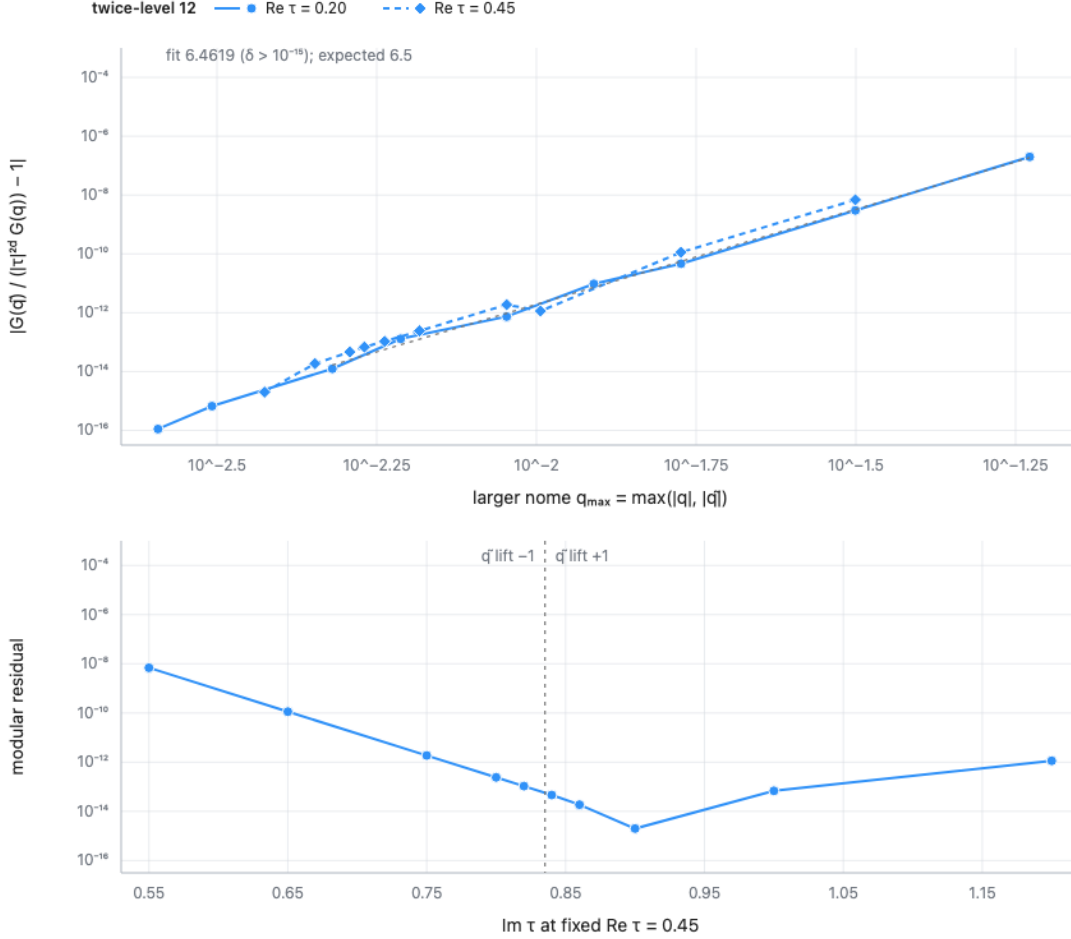


Figure 2: Cannon scan of the genus-one NS one-point modular residual at the highest recursion cutoff, twice-level 12. Solid circles and dashed diamonds are the $\text{Re } \tau = 0.20$ and 0.45 families. The upper panel shows the residual against the larger nome; the lower panel resolves the transformed NS lift change.

This completes the scalar genus-one one-point function with a bottom NS external primary in both even-spin modular orbits; it is not yet the entire physical torus amplitude. For multipuncture R sewing and higher genus, the next representation-theory step is to keep the 2×2 R ground indices open at every positive level and recover the scalar Eq. (144) only after a cycle projection. The starred external component, the short quotient at $P = 0$, and the Berezin orientation selecting the physical G_0 insertion remain explicit separate tasks.

For the higher-genus c -recursion the same operation should be performed algebraically: store each coefficient as a Laurent/partial-fraction object in c , merge coincident pole keys, and evaluate at c_0 only after the recursive sum is complete. Equation (136) is the sphere implementation and

numerical oracle for that graph-level partial-fraction engine.

8 Edge-local residue derivation

First fix an internal NS edge e of weight h_e . At $c = c_{rs}(h_e)$, the inverse Gram matrix has a singular rank-one block along the null submodule. For descendants K, L above the null state,

$$\begin{aligned} \text{Res}_{c=c_{rs}(h_e)} (B_{h_e, c}^{(rs/2+n)})^{-1} [\mathcal{L}_K \chi_{rs}, \mathcal{L}_L \chi_{rs}] \\ = J_{rs}(h_e) A_{rs}(h_e) (B_{h_e + rs/2, c_{rs}}^{(n)})^{-1} [K, L]. \end{aligned}$$

The two vertex tensors adjacent to e factor through Eqs. (38)–(39). Every other edge and vertex is a spectator. Therefore

$$\text{Res}_{c=c_{rs}(h_e)} \mathcal{F}_\Gamma^\alpha = q_e^{rs/2} \mathcal{R}_{rs}^{(e)} \mathcal{F}_\Gamma^\alpha \left(c_{rs}(h_e), \mathbf{h} + \frac{rs}{2} \mathbf{e}_e; \mathbf{q} \right), \quad (176)$$

where

$$\mathcal{R}_{rs}^{(e)} = J_{rs}(h_e) A_{rs}(h_e) \Sigma_{rs}^{(v_L, e)} \Sigma_{rs}^{(v_R, e)}. \quad (177)$$

Each Σ is the σ or $S\sigma$ prescribed by the ordered local tensor. Its two weight arguments are the other two legs at that vertex. For an NS edge this is precisely the CCY locality mechanism with the replacements

$$rs \longrightarrow \frac{rs}{2}, \quad P_{rs} \longrightarrow \sigma_{rs}^\alpha, \quad \mathfrak{sl}(2) \longrightarrow \mathfrak{osp}(1|2). \quad (178)$$

Now let e be a generic long Ramond edge and retain its ground indices. The inverse-Gram residue is a map on the ground fiber and factorizes through the shifted long module. Put $c_* = c_{rs}^R(\beta_e; \epsilon)$. Then

$$\text{Res}_{c=c_*} (B_{R, \beta_e, c}^{(rs/2+n)})^{-1} = J_{rs}^R(\beta_e; \epsilon) A_{rs}^{R, (h)} D_{rs} (B_{R, \beta'_{rs}, c_*}^{(n)})^{-1} D_{rs}^{\text{BPZ}}, \quad r+s \text{ odd}. \quad (179)$$

Here the displayed equation is understood as the singular block between descendants of the two null partners; D_{rs} includes their fixed relative G_0 phase. Applying the fusion maps at the adjacent vertices gives

$$\text{Res}_{c=c_*} \mathbf{F}_\Gamma = q_e^{rs/2} J_{rs}^R(\beta_e; \epsilon) A_{rs}^{R, (h)} \mathbf{P}_{rs}^{(v_L, e; \epsilon)} \mathbf{F}_\Gamma(c_*, \mathbf{h}_{\text{NS}}, \beta_R + (\beta'_{rs} - \beta_e) \mathbf{e}_e; \mathbf{q}) \mathbf{P}_{rs}^{(v_R, e; \epsilon)}. \quad (180)$$

The order in Eq. (180) is part of the formula. It follows the oriented Ramond defect line; exchanging the two matrices can change the answer. All nonadjacent graph data are spectators, exactly as in the NS derivation.

At generic weights the pole positions contributed by different edges are distinct and all poles are simple. At coincident or minimal-model values, the implementation must merge equal pole keys and carry the resulting higher Laurent orders. The self-dual finite-part prescription Eq. (136) is the tested sphere realization of this rule. A one-sided numerical regulator is useful only as an independent check.

9 Genus- g NS/R c -recursion

Use the mixed analytic coordinates

$$\mathbf{F}_\Gamma = \mathbf{F}_\Gamma(c; \{h_e\}_{e \in E_{\text{NS}}}, \{\beta_e\}_{e \in E_{\text{R}}}; \mathbf{q}). \quad (181)$$

Thus NS weights and Ramond momenta, rather than Ramond absolute weights, are held fixed as c varies. Assume that the corresponding reduced plumbing-frame block has a controlled $c \rightarrow \infty$ limit, and keep every open R ground index. For each Ramond summand abbreviate

$$\begin{aligned} c_{e,rs,\epsilon}^* &= c_{rs}^R(\beta_e; \epsilon), & \beta_R^* &= \beta_R + (\beta'_{rs}(b_*) - \beta_e)e_e, \\ \mathcal{K}_{e,rs,\epsilon}^R(c) &= \frac{q_e^{rs/2} J_{rs}^R(\beta_e; \epsilon) A_{rs}^{R,(h)}(c_{e,rs,\epsilon}^*)}{c - c_{e,rs,\epsilon}^*}. \end{aligned} \quad (182)$$

The positive-level polar part is

$$\begin{aligned} \mathbf{F}_\Gamma^\alpha(c, \mathbf{h}_{\text{NS}}, \beta_R; \mathbf{q}) &= \mathbf{F}_{\Gamma,\infty}^\alpha(\mathbf{h}_{\text{NS}}, \beta_R; \mathbf{q}) \\ &+ \sum_{e \in E_{\text{NS}}} \sum_{\substack{r \geq 2, s \geq 1 \\ r+s \in 2\mathbb{Z}}} \frac{q_e^{rs/2} J_{rs}(h_e) A_{rs}(h_e)}{c - c_{rs}(h_e)} \Sigma_{rs}^{(v_L, e)} \Sigma_{rs}^{(v_R, e)} \mathbf{F}_\Gamma^\alpha \left(c_{rs}(h_e), \mathbf{h}_{\text{NS}} + \frac{rs}{2} \mathbf{e}_e, \beta_R; \mathbf{q} \right). \end{aligned} \quad (183)$$

$$\begin{aligned} \mathbf{F}_\Gamma^\alpha|_{\text{R poles}} &= \sum_{\substack{e \in E_R, r \geq 2, s \geq 1 \\ r+s \in 2\mathbb{Z}+1, \epsilon = \pm 1}} \mathcal{K}_{e,rs,\epsilon}^R(c) \mathbf{P}_{rs}^{(v_L, e; \epsilon)} \\ &\times \mathbf{F}_\Gamma^\alpha(c_{e,rs,\epsilon}^*, \mathbf{h}_{\text{NS}}, \beta_R^*; \mathbf{q}) \mathbf{P}_{rs}^{(v_R, e; \epsilon)}. \end{aligned} \quad (184)$$

At fixed multi-level $\mathbf{n}$, the sum on edge e is finite: $rs \leq 2n_e$. Hence Eqs. (183) and (184) give a triangular coefficient recursion. The two-dimensional R matrix index is transported through the recursion; it is not an extra summation over representations. Choosing a particular R ground state or GSO component is a boundary contraction performed after the generic long block has been computed. Equal c_* values produced by different quadratic branches are merged before evaluation.

At fixed β_e , the ground denominator is

$$\frac{1}{h_e^R - c/24} = -\frac{1}{\beta_e^2}, \quad (185)$$

so there is no additional level-zero c -pole for generic $\beta_e \neq 0$. The short module at $\beta_e = 0$ remains a separate irreducible quotient and is taken only after the generic long block is constructed.

The formula should be implemented as a recursion on a decorated graph, not as a new closed expression for each topology. The topology-dependent data are only:

$$(\Gamma, s_e, \text{half-edge orderings}, \alpha_v, q_e, \text{spin lift}, \text{R-defect orientation}). \quad (186)$$

The NS kernel $(c_{rs}, J_{rs}, A_{rs}, \sigma_{rs}^\alpha)$ and the R kernel $(c_{rs}^R(\beta; \epsilon), J_{rs}^R(\beta; \epsilon), A_{rs}^{R,(h)}, \mathbf{P}_{rs}^{(\epsilon)})$ are universal.

For the theta graph, an NS pole on edge $e = i$ shifts only h_i , whereas a Ramond pole shifts only $\beta_i \rightarrow \beta'_{rs}$. In both cases the shifted theta block is multiplied by the two fusion factors at the left and right vertices. For an NS edge,

$$\mathcal{R}_{rs}^{(i)} = J_{rs}(h_i) A_{rs}(h_i) \Sigma_{rs}^{(L,i)}(h_j, h_k; h_i) \Sigma_{rs}^{(R,i)}(h_j, h_k; h_i), \quad \{i, j, k\} = \{1, 2, 3\}. \quad (187)$$

The displayed Σ notation is intentional: the sign depends on which slot of each ordered three-point tensor contains edge i . The graph object, not a handwritten topology formula, is the single source of truth.

10 The large- c regular part

10.1 All-NS super-CCY factorization

For an all-NS graph, split each descendant into global and non-global modes. The global subalgebra is generated by

$$L_{0,\pm 1}, \quad G_{\pm 1/2}, \quad (188)$$

and is $\mathfrak{osp}(1|2)$. The remaining modes are L_{-n} with $n \geq 2$ and G_{-r} with $r \geq 3/2$. At fixed weights, Gram-Schmidt can be used to form a basis whose norms have definite powers of c . Super-Virasoro Ward identities then force the non-global modes in a leading three-point tensor to contract among themselves through their central terms. Dependence on the finite h_i is subleading in these contractions. What remains is a product of:

1. the same non-global-mode tensor with all modules replaced by the NS vacuum module;
2. a three-point tensor constructed only from L_{-1} and $G_{-1/2}$ descendants.

The inverse Gram matrices factorize in the same way. Repeating this at every vertex yields the super-analogue of the CCY result,

$$\boxed{\mathcal{F}_{\Gamma,\infty}^\alpha(\mathbf{h}; \mathbf{q}, \delta) = \mathcal{V}_{\Gamma,NS}^{(\infty)}(\mathbf{q}; \delta) \mathcal{G}_\Gamma^{\mathfrak{osp}(1|2),\alpha}(\mathbf{h}; \mathbf{q}, \delta).} \quad (189)$$

The spin structure δ enters through the lift of fermionic sewing around cycles. Equation (189) is the correct seed organization; it is also the part of the argument that must receive direct low-level checks at genus two.

The global factor is finite-dimensional representation theory: retain only $L_{-1}^n \nu_h$ and $L_{-1}^n G_{-1/2} \nu_h$ on every NS edge and contract their $\mathfrak{osp}(1|2)$ Gram matrices and vertex tensors on Γ . This is a sparse sum over (n_e, β_e) with $\beta_e \in \{0, 1\}$.

For a graph with Ramond edges, the polar kernel derived above is unchanged, but one should not reuse Eq. (189) without a separate large- c analysis. Integer fermion modes, the G_0 ground fiber, and the Ramond sewing insertion alter the regular seed. The safe definition is again direct sewing at large c in the same ordered ground basis. Thus the representation-theory residue problem and the R large- c seed problem are logically separate.

10.2 Vacuum seed: known checks and the remaining problem

We define the higher-genus vacuum seed without assuming a closed product:

$$\mathcal{V}_{\Gamma,NS}^{(\infty)}(\mathbf{q}; \delta) \equiv \lim_{c \rightarrow \infty} \mathcal{F}_\Gamma^{\text{vac},\delta}(c; \mathbf{q}), \quad (190)$$

where the right-hand side is evaluated by Eq. (19) with the NS vacuum module on every internal and external leg. This definition fixes all normalization and signs in the same plumbing frame as the recursion.

At genus one and even NS spin structure it reduces, after removing the Casimir factor, to the vacuum oscillator character

$$\mathcal{V}_{1,NS}^{(\infty)}(q) = \prod_{n=2}^{\infty} \frac{1 + q^{n-1/2}}{1 - q^n}. \quad (191)$$

In the bosonic problem, CCY identify the higher-genus vacuum seed, in their holomorphic plumbing normalization, with

$$\mathcal{V}_{\Gamma, \text{Vir}}^{(\infty)} = \prod_{\gamma \in \mathcal{P}} \prod_{n=2}^{\infty} (1 - k_{\gamma}^n)^{-1/2}, \quad (192)$$

where $\mathcal{P}$ is the set of primitive Schottky conjugacy classes and k_{γ} is the multiplier. A supersymmetric analogue should contain a lift of each Schottky element to the spin bundle, and hence a sign depending on δ . We do not assume a particular fermionic product or exponent in this note. The safe sequence is:

$$\begin{aligned} \text{direct vacuum sewing} &\longrightarrow \text{low-order series in } q_e \\ &\longrightarrow \text{candidate super-Schottky product} \\ &\longrightarrow \text{coefficient-by-coefficient proof/test.} \end{aligned} \quad (193)$$

This seed, rather than the degenerate pole algebra, is the main unresolved higher-genus ingredient.

11 Supergeometry and the sewing ledger

The algebraic block does not by itself specify a super-Riemann surface. For an NS node, the reduced bosonic sewing parameter is a square of a superconformal gluing parameter, schematically

$$q_e = -\varepsilon_e^2. \quad (194)$$

Because NS descendants occur at half-integer level, choosing q_e without a choice of ε_e loses the sign of $q_e^{1/2}$. A numerical block object must therefore store a lifted parameter or an equivalent spin sign, not just a complex number q_e . This is also why changing branches inside a recursion call can spoil crossing or modular covariance.

The dimension of supermoduli space is

$$\dim \mathfrak{M}_{g, n_{\text{NS}}, n_{\text{R}}} = \left(3g - 3 + n_{\text{NS}} + n_{\text{R}} \mid 2g - 2 + n_{\text{NS}} + \frac{n_{\text{R}}}{2} \right). \quad (195)$$

In particular, a three-NS-punctured sphere has one odd modulus. The two kinematic forms ρ^0 and ρ^1 are the component-level shadow of this fact. A physical string amplitude chooses or integrates the corresponding odd directions through its measure and picture prescription; a chiral block engine must preserve the label rather than summing it prematurely.

For every plumbing graph we will keep a separate ledger containing:

edge data:	NS/R, h_e , q_e , lift/sign, orientation,	(196)
vertex data:	cyclic half-edge order, α_v , odd-modulus convention,	
global data:	δ , homology basis, defect-line routing,	
frame data:	local coordinates and anomaly normalization.	

The super-Riemann-surface formalism of Ref. [7] supplies these data. The recursion kernel consumes them.

12 Ramond shortening and the geometric sewing boundary

The representation-theory part of a generic long-R edge is now contained in Eqs. (16)–(73). No new Kac mechanism appears at higher genus. What remains genuinely geometric is the map that contracts the ground fiber across a Ramond node.

Ramond gluing carries an odd plumbing datum. In an operator description its Berezin integral selects a prescribed combination of the identity and G_0 on the two-dimensional ground fiber. That insertion changes the final ground-index contraction, but not the positive-level pole locations, null-norm slopes, or fusion polynomials. It should therefore be represented as a sewing matrix attached to the edge and applied after Eq. (184). Its sign depends on the orientation of the odd measure.

There are two further boundary choices:

1. At $h = c/24$, replace the long module by its irreducible short quotient. The normalized two-state basis is singular there, while the unnormalized basis (17) has a regular algebraic limit followed by an explicit projection.
2. In type 0B, the $V^{\mathbf{R},-}$ family is attached to the fermion-parity topological line. The R ground matrices must be multiplied along an oriented defect path or cycle, with junction and GSO phases supplied by the sewing ledger.

These choices do not obstruct the generic recursion; they select a component of the matrix-valued block. The sphere and torus R recursions fix the local matrix convention, while the super-Riemann-surface formalism fixes its higher-genus contraction.

13 Numerical architecture

The recommended implementation has six layers.

Layer	Responsibility
Module algebra	NS bases and the unnormalized long-R ground doublet, fermionic ordering, Gram matrices, and direct low-level null vectors.
Null kernel	Sector-specific Kac labels, branch-tracked c_{rs} , J_{rs} , inverse norm slopes A_{rs} , and shifted-module parameters.
Vertex kernel	NS fusion scalars $\sigma_{rs}^{0,1}$, R fusion matrices $\mathbf{P}_{rs}$, Ward-identity phases, and canonical slot orderings.
Meromorphic ledger	Laurent/partial-fraction coefficients in c , merged coincident pole keys, higher pole orders, and finite-part evaluation only after a complete recursive coefficient has been assembled.
Decorated graph	Topology, half-edge order, component labels, edge sectors, plumbing lifts, R-defect orientation, and sparse multi-level truncation.
Seed and recursion	Sector-appropriate large- c seed, memoized edge-local recursion, ground-index contractions, and numerical error ledger.

A coefficient evaluator at total twice-level cutoff L should:

1. enumerate multi-levels $\ell_e = 2n_e \in \mathbb{Z}_{\geq 0}$ with $\sum_e \ell_e \leq L$;
2. evaluate the large- c seed coefficient;
3. for each NS edge enumerate $r + s$ even and for each R edge enumerate $r + s$ odd, always with $r \geq 2$, $s \geq 1$, and $rs \leq \ell_e$;

4. multiply $J_{rs}A_{rs}$ by the two canonicalized NS fusion scalars or by the ordered pair of R fusion matrices;
5. recurse at the sector-specific $c_{rs}(h_e)$ with $h_e \mapsto h_e + rs/2$ and $\ell_e \mapsto \ell_e - rs$;
6. memoize by the full shifted weight tuple, central charge, multi-level, component tuple, R ground indices, and branch/lift identifiers.

Memoizing only numerical q_e and weights is insufficient because two spin lifts can have the same reduced q_e .

14 Validation ladder

The implementation should not begin with a physical genus-two integral. The following ladder isolates errors cheaply.

1. **Algebra.** Reproduce $\langle G_{-1/2}\nu_h | G_{-1/2}\nu_h \rangle = 2h$ and the direct Gram and three-point tensors through at least total level 2. In the R module verify $G_0^2 = h - c/24$, $\det B_R^{(0)} = h - c/24$, and the level-one Kac zero.
2. **NS sphere.** Recover every coefficient produced by the existing `superconformal_blocks.py` layer, including starred external components and the S_{rs}^α sign.
3. **Four-R sphere.** With NS exchange, reproduce Eqs. (78)–(80) for all four HJS chiral sign branches. This calibrates $P_{\text{NS}|\text{RR}}^{rs}$ and the odd–odd phase, but not an R internal edge.
4. **R-internal sphere.** At generic complex β , use a two-NS/two-R ordering with R exchange to reproduce the two $\pm\beta_{rs}$ residues and verify that converting them to a weight pole gives Eqs. (67)–(69). Reproduce the direct descendant coefficients through at least q^3 , as in Sec. 4.7, and keep all ground indices open before any nonchiral contraction.
5. **Torus.** Recover Eq. (191), the published NS one-point recursions, and the generic long-R toric recursion including the shift $\beta_{rs} \mapsto \beta'_{rs}$.
6. **Residue.** For a generic complex weight, compare a numerical finite-difference residue of direct sewing with Eqs. (176), (177), and (180).
7. **All-NS genus two.** Compare recursion and direct theta-graph sewing coefficient by coefficient at total twice-level $L = 2, 4, 6$ before increasing the cutoff.
8. **Mixed genus.** Choose the smallest graph containing an oriented R cycle, keep its ground indices open, and compare direct sewing with the ordered matrix recursion before applying the odd-plumbing or GSO projector.
9. **Geometry.** Flip one plumbing lift while holding q_e fixed and verify the predicted sign of half-integer-level terms. Then check graph automorphisms and a basic change of pants channel.
10. **Shortening.** Approach $\beta = 0$ in the unnormalized basis, project out $G_0 w^+$, and compare with a direct short-module computation.
11. **Seed.** Only after the direct vacuum coefficients are stable, test a proposed super-Schottky primitive-class product.

For the type-0B application, generic complex c remains useful while debugging individual residues and branches. Production coefficients at $c = 27/2$ should instead be assembled as Laurent/partial-fraction objects and evaluated only after coincident poles have been merged. On the sphere, Eq. (136) implements this rule directly; the old $c_0 \pm \epsilon$ average is retained as an independent regression.

15 Status and immediate next calculation

Ingredient	Status	Next action
Ordinary- c NS pole locations and Jacobian	Fixed	Reuse the tested sphere kernel.
RRRR/NSNSRR fixed-weight c -pole experiment	Retained as a local residue test only	First $(3, 1)$ and $(2, 2)$ residues agree with the independent h -recursion. Replace its analytic slice by fixed Ramond β_i ; do not reuse its fixed- h_i^R large- c seed.
NS null norm and two fusion families	Fixed	Refactor into a topology-free module with canonical orientations.
Four-R sphere block with NS exchange	Implemented and tested	Reuse the fixed HJS-to-BRY sign map in later nonchiral contractions.
Exact $b = 1$ collision handling	Implemented for sphere coefficients	Promote the tested finite-part oracle to graph-level partial fractions in c .
RRRR physical crossing	Implemented and tested	Retain as a regression for the HJS-to-BRY chiral sign dictionary.
Generic long-R sphere block	Implemented and tested	Level-one direct sewing, direct descendants through q^3 , exact- c finite part, and RRNSNS mixed-sector crossing pass.
NS/R torus one-point kernel	Exact $b = 1$, $2n_{\max} = 12$; S error 1.6×10^{-15}	Open the positive-level R ground indices before the cycle projection; then test the $\widetilde{\text{NS}} \leftrightarrow \text{R}$ orbit.
NS torus two-point necklace	Two-edge h -recursion and $dP_a dP_b / \pi^2$ assembly implemented	Extend the direct Ward/Gram oracle beyond level one and run production four-parameter convergence scans.
R-handle NS two-point necklace	Paired beta-pole recursion, all sign/cycle sectors, exact- $b = 1$ finite part, and spectral assembly through level one	Derive the contracted all-level R regular seed and replace the direct Ward/Gram seed oracle.
R null norm and fusion matrices	Sphere scalar components calibrated	Lift the tested local residue to open ground-fiber matrices for graph sewing.
Arbitrary-graph NS/R polar recursion	Derived	Implement sector-decorated edges and ordered matrix insertions.
$\mathfrak{osp}(1 2)$ global factor	Fixed in principle	Implement sparse global-mode contractions for the all-NS graph.
Higher-genus NS vacuum seed	Open but sharply defined	Compute the theta vacuum series directly through low total level; only then identify a closed Schottky formula.
R large- c seed and odd sewing matrix	Open but separated	Obtain by direct long-R sewing in a fixed Berezin convention.
Spin structure and defect routing	Ledger specified ⁴²	Fix one genus-two homology, plumbing, and oriented R-cycle convention.
Short R ground state	Boundary case isolated	Project the unnormalized long module at $\beta = 0$ after generic tests pass.

The sphere calibration now includes four-R/NS exchange through order 12 at the exact self-dual point, the generic long-R direct Gram/Ward calculation through q^3 , and physical RRRR and RRNSNS crossing. The genus-one layer adds the NS spin lift, both scalar HJS R projections, and an explicit G_0 ground-fiber closure. The local G_0 action and scalar R residue are therefore fixed through three descendant levels. The next calculation is to keep both Ramond ground indices open at every positive level and promote the tested scalar factors to the ordered matrices $J_{rs}^R(\beta; \epsilon) A_{rs}^{R, (h)} \mathbf{P}_L^{(\epsilon)} \mathbf{P}_R^{(\epsilon)}$. Every component should first be matched to direct two-NS/two-R sewing and to the published toric block. This matching and the subsequent large- c analysis must keep every Ramond β fixed. Only then should the matrix be inserted on an oriented R cycle of a genus-two graph. In parallel, the all-NS theta block at total twice-level $L = 2$ and 4 should be compared between direct sewing and Eq. (183), with coincident c poles stored in a merged partial-fraction ledger. These tests separate the remaining ground-fiber routing and orientation choices from the higher-genus seed and sewing ledger.

16 Conclusion

The CCY locality argument extends to both NS and generic long-R super-Virasoro modules. In either sector the polar coefficient is exactly $J_{rs} A_{rs}$ times one null-vector fusion factor at each endpoint. The R sector replaces scalar endpoint data by ordered maps on a two-dimensional G_0 ground fiber; the fiber dimension is unchanged by the null-module shift. Its natural c -plane coordinate keeps β , rather than h_R , fixed. Hence the genus- g polar recursion is not a new representation problem: it is graph assembly of the sphere-tested local kernel.

The genuinely new higher-genus work lies in the regular large- c seed and in the supergeometric contraction data. Spin lifts, odd Ramond plumbing, and defect routing select components of the algebraic matrix block but do not modify its positive-level Kac residues. The short R ground state is a controlled quotient at $\beta = 0$, not a new generic recursion. This separation gives a precise route from known four-point representation theory to a numerical genus- g block engine.

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